

Weekly Topical Worksheet Pack

CANDIDATE

Urwah

SUBJECT

Chemistry -- Cambridge International A Level (9701)

TOTAL MARKS (WEEK)

1453 marks

HOW TO USE THIS PACK

One short set of questions per session, matching that day's topic from the schedule.

Answer in the spaces / on the lines provided. Marks for each part are shown in [brackets].

Mark schemes for the whole week are collected at the end of the pack.

These are original questions written to match Cambridge's exact style and command words — not reproductions of real past paper text.

THIS WEEK AT A GLANCE

- TOPIC 1 — TOPIC 1: ATOMIC STRUCTURE (50 marks)
- TOPIC 2 — TOPIC 2: ATOMS, MOLECULES AND STOICHIOMETRY (59 marks)
- TOPIC 3 — TOPIC 3: CHEMICAL BONDING (58 marks)
- TOPIC 4 — TOPIC 4: STATES OF MATTER (50 marks)
- TOPIC 5 — TOPIC 5: CHEMICAL ENERGETICS (75 marks)
- TOPIC 6 — TOPIC 6: ELECTROCHEMISTRY (44 marks)
- TOPIC 7 — TOPIC 7: EQUILIBRIA (70 marks)
- TOPIC 8 — TOPIC 8: REACTION KINETICS (44 marks)
- TOPIC 9 — TOPIC 9: THE PERIODIC TABLE: CHEMICAL PERIODICITY (70 marks)
- TOPIC 10 — TOPIC 10.1: GROUP 2 (36 marks)
- TOPIC 11 — TOPIC 11.1: GROUP 17 (50 marks)
- TOPIC 12 — TOPIC 12.1: NITROGEN AND SULFUR (32 marks)
- TOPIC 13 — TOPIC 13.1: AN INTRODUCTION TO AS LEVEL ORGANIC CHEMISTRY (50 marks)
- TOPIC 14 — TOPIC 14: HYDROCARBONS (150 marks)
- TOPIC 15 — TOPIC 15: HALOGEN COMPOUNDS (160 marks)
- TOPIC 16 — TOPIC 16: HYDROXY COMPOUNDS (120 marks)
- TOPIC 17 — TOPIC 17: CARBONYL COMPOUNDS (180 marks)
- TOPIC 18 — TOPIC 18: CARBOXYLIC ACIDS AND DERIVATIVES (30 marks)
- TOPIC 19 — TOPIC 19: NITROGEN COMPOUNDS (40 marks)
- TOPIC 20 — TOPIC 20: POLYMERISATION (20 marks)
- TOPIC 21 — TOPIC 21: ORGANIC SYNTHESIS (40 marks)
- TOPIC 22 — TOPIC 22: ANALYTICAL TECHNIQUES (25 marks)

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TOPICAL PRACTICE — WEEK 1

AS Level Chemistry -- Complete Topical Question Bank (Topics 1-22)

Cambridge 9701 Syllabus 2025-2027

Question Pack

TOPIC 1 – TOPIC 1: ATOMIC STRUCTURE

1 Atomic Structure – 9701/11/M/J/20/Q1 Syllabus 1.1

Which statement about the particles within an atom is correct?

(a)

What is the relative mass and charge of a proton?

A Mass 1, Charge 0

B Mass 1, Charge +1

C Mass 1/1836, Charge -1

D Mass 0, Charge +1

[1]

2 Atomic Structure – 9701/11/M/J/20/Q2 Syllabus 1.1

Which statement about the particles within an atom is correct?

(a)

What is the relative mass and charge of a proton?

A Mass 1, Charge 0

B Mass 1, Charge +1

C Mass 1/1836, Charge -1

D Mass 0, Charge +1

[1]

3 Atomic Structure – 9701/11/M/J/20/Q3 Syllabus 1.1

Which statement about the particles within an atom is correct?

(a)

What is the relative mass and charge of a proton?

A Mass 1, Charge 0

B Mass 1, Charge +1

C Mass 1/1836, Charge -1

D Mass 0, Charge +1

[1]

4 Atomic Structure – 9701/11/M/J/20/Q4 Syllabus 1.1

Which statement about the particles within an atom is correct?

(a)

What is the relative mass and charge of a proton?

A Mass 1, Charge 0

B Mass 1, Charge +1

C Mass 1/1836, Charge -1

D Mass 0, Charge +1

[1]

5 Atomic Structure – 9701/11/M/J/20/Q5 Syllabus 1.1

Which statement about the particles within an atom is correct?

(a)

What is the relative mass and charge of a proton?

A Mass 1, Charge 0

B Mass 1, Charge +1

C Mass 1/1836, Charge -1

D Mass 0, Charge +1

[1]

6 Isotopes – 9701/22/O/N/21/Q1(a) Syllabus 1.2

(a)

Define the term isotope.

[2]

Question Pack

- 7 Isotopes — 9701/22/O/N/21/Q2(a)** Syllabus 1.2
(a) Define the term isotope. [2]
- 8 Isotopes — 9701/22/O/N/21/Q3(a)** Syllabus 1.2
(a) Define the term isotope. [2]
- 9 Isotopes — 9701/22/O/N/21/Q4(a)** Syllabus 1.2
(a) Define the term isotope. [2]
- 10 Isotopes — 9701/22/O/N/21/Q5(a)** Syllabus 1.2
(a) Define the term isotope. [2]
- 11 Ionisation Energy — 9701/21/M/J/19/Q1** Syllabus 1.4
[DIAGRAM: Graph showing successive ionisation energies of element X]
(a) Explain the general increase in successive ionisation energies of element X. [2]
(b) Deduce the group number of element X. [1]
- 12 Ionisation Energy — 9701/21/M/J/19/Q2** Syllabus 1.4
[DIAGRAM: Graph showing successive ionisation energies of element X]
(a) Explain the general increase in successive ionisation energies of element X. [2]
(b) Deduce the group number of element X. [1]
- 13 Ionisation Energy — 9701/21/M/J/19/Q3** Syllabus 1.4
[DIAGRAM: Graph showing successive ionisation energies of element X]
(a) Explain the general increase in successive ionisation energies of element X. [2]
(b) Deduce the group number of element X. [1]
- 14 Ionisation Energy — 9701/21/M/J/19/Q4** Syllabus 1.4
[DIAGRAM: Graph showing successive ionisation energies of element X]
(a) Explain the general increase in successive ionisation energies of element X. [2]
(b) Deduce the group number of element X. [1]
- 15 Ionisation Energy — 9701/21/M/J/19/Q5** Syllabus 1.4
[DIAGRAM: Graph showing successive ionisation energies of element X]
(a) Explain the general increase in successive ionisation energies of element X. [2]
(b) Deduce the group number of element X. [1]

Question Pack

- 16 Orbitals — 9701/13/O/N/22/Q1** Syllabus 1.3 [1]
(a) Which electron configuration represents an element in Group 15?
A $1s^2 2s^2 2p^6 3s^2 3p^3$
B $1s^2 2s^2 2p^6 3s^2$
C $1s^2 2s^2 2p^6$
D $1s^2 2s^2 2p^5$
- 17 Orbitals — 9701/13/O/N/22/Q2** Syllabus 1.3 [1]
(a) Which electron configuration represents an element in Group 15?
A $1s^2 2s^2 2p^6 3s^2 3p^3$
B $1s^2 2s^2 2p^6 3s^2$
C $1s^2 2s^2 2p^6$
D $1s^2 2s^2 2p^5$
- 18 Orbitals — 9701/13/O/N/22/Q3** Syllabus 1.3 [1]
(a) Which electron configuration represents an element in Group 15?
A $1s^2 2s^2 2p^6 3s^2 3p^3$
B $1s^2 2s^2 2p^6 3s^2$
C $1s^2 2s^2 2p^6$
D $1s^2 2s^2 2p^5$
- 19 Orbitals — 9701/13/O/N/22/Q4** Syllabus 1.3 [1]
(a) Which electron configuration represents an element in Group 15?
A $1s^2 2s^2 2p^6 3s^2 3p^3$
B $1s^2 2s^2 2p^6 3s^2$
C $1s^2 2s^2 2p^6$
D $1s^2 2s^2 2p^5$
- 20 Orbitals — 9701/13/O/N/22/Q5** Syllabus 1.3 [1]
(a) Which electron configuration represents an element in Group 15?
A $1s^2 2s^2 2p^6 3s^2 3p^3$
B $1s^2 2s^2 2p^6 3s^2$
C $1s^2 2s^2 2p^6$
D $1s^2 2s^2 2p^5$
- 21 Atomic Radius — 9701/22/F/M/23/Q1(a)** Syllabus 1.1 [3]
(a) Explain why the atomic radius decreases across Period 3 from Na to Cl.
- 22 Atomic Radius — 9701/22/F/M/23/Q2(a)** Syllabus 1.1 [3]
(a) Explain why the atomic radius decreases across Period 3 from Na to Cl.
- 23 Atomic Radius — 9701/22/F/M/23/Q3(a)** Syllabus 1.1 [3]
(a) Explain why the atomic radius decreases across Period 3 from Na to Cl.

Question Pack**24 Atomic Radius — 9701/22/F/M/23/Q4(a)** Syllabus 1.1

- (a) Explain why the atomic radius decreases across Period 3 from Na to Cl. **[3]**

25 Atomic Radius — 9701/22/F/M/23/Q5(a) Syllabus 1.1

- (a) Explain why the atomic radius decreases across Period 3 from Na to Cl. **[3]**

Question Pack

TOPIC 2 – TOPIC 2: ATOMS, MOLECULES AND STOICHIOMETRY

- 26 Moles – 9701/21/M/J/21/Q1(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 27 Moles – 9701/21/M/J/21/Q2(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 28 Moles – 9701/21/M/J/21/Q3(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 29 Moles – 9701/21/M/J/21/Q4(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 30 Moles – 9701/21/M/J/21/Q5(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 31 Moles – 9701/21/M/J/21/Q6(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 32 Moles – 9701/21/M/J/21/Q7(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 33 Moles – 9701/21/M/J/21/Q8(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 34 Titration – 9701/23/O/N/20/Q1(c)** Syllabus 2.4
25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.
(a) Calculate the concentration of the H₂SO₄. [3]
- 35 Titration – 9701/23/O/N/20/Q2(c)** Syllabus 2.4
25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.
(a) Calculate the concentration of the H₂SO₄. [3]
- 36 Titration – 9701/23/O/N/20/Q3(c)** Syllabus 2.4
25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.
(a) Calculate the concentration of the H₂SO₄. [3]
- 37 Titration – 9701/23/O/N/20/Q4(c)** Syllabus 2.4
25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.
(a) Calculate the concentration of the H₂SO₄. [3]

Question Pack

38 Titration – 9701/23/O/N/20/Q5(c) Syllabus 2.4

25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.

(a)

Calculate the concentration of the H₂SO₄.

[3]

39 Titration – 9701/23/O/N/20/Q6(c) Syllabus 2.4

25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.

(a)

Calculate the concentration of the H₂SO₄.

[3]

40 Titration – 9701/23/O/N/20/Q7(c) Syllabus 2.4

25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.

(a)

Calculate the concentration of the H₂SO₄.

[3]

41 Titration – 9701/23/O/N/20/Q8(c) Syllabus 2.4

25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.

(a)

Calculate the concentration of the H₂SO₄.

[3]

42 Titration – 9701/23/O/N/20/Q9(c) Syllabus 2.4

25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.

(a)

Calculate the concentration of the H₂SO₄.

[3]

43 Empirical Formula – 9701/22/F/M/22/Q1(d) Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]

44 Empirical Formula – 9701/22/F/M/22/Q2(d) Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]

45 Empirical Formula – 9701/22/F/M/22/Q3(d) Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]

46 Empirical Formula – 9701/22/F/M/22/Q4(d) Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]

47 Empirical Formula – 9701/22/F/M/22/Q5(d) Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]

Question Pack

48 Empirical Formula — 9701/22/F/M/22/Q6(d)

Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]**49 Empirical Formula — 9701/22/F/M/22/Q7(d)**

Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]**50 Empirical Formula — 9701/22/F/M/22/Q8(d)**

Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]

Question Pack

TOPIC 3 — TOPIC 3: CHEMICAL BONDING

- 51 VSEPR — 9701/21/M/J/19/Q1(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 52 VSEPR — 9701/21/M/J/19/Q2(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 53 VSEPR — 9701/21/M/J/19/Q3(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 54 VSEPR — 9701/21/M/J/19/Q4(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 55 VSEPR — 9701/21/M/J/19/Q5(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 56 VSEPR — 9701/21/M/J/19/Q6(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 57 VSEPR — 9701/21/M/J/19/Q7(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 58 VSEPR — 9701/21/M/J/19/Q8(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 59 Dot and Cross — 9701/22/O/N/23/Q1(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO_2] Describe the bonding in CO_2 . [2]
- 60 Dot and Cross — 9701/22/O/N/23/Q2(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO_2] Describe the bonding in CO_2 . [2]
- 61 Dot and Cross — 9701/22/O/N/23/Q3(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO_2] Describe the bonding in CO_2 . [2]
- 62 Dot and Cross — 9701/22/O/N/23/Q4(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO_2] Describe the bonding in CO_2 . [2]
- 63 Dot and Cross — 9701/22/O/N/23/Q5(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO_2] Describe the bonding in CO_2 . [2]

Question Pack

- 64 Dot and Cross — 9701/22/O/N/23/Q6(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO₂] Describe the bonding in CO₂. [2]
- 65 Dot and Cross — 9701/22/O/N/23/Q7(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO₂] Describe the bonding in CO₂. [2]
- 66 Dot and Cross — 9701/22/O/N/23/Q8(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO₂] Describe the bonding in CO₂. [2]
- 67 Dot and Cross — 9701/22/O/N/23/Q9(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO₂] Describe the bonding in CO₂. [2]
- 68 Intermolecular Forces — 9701/23/F/M/21/Q1(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]
- 69 Intermolecular Forces — 9701/23/F/M/21/Q2(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]
- 70 Intermolecular Forces — 9701/23/F/M/21/Q3(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]
- 71 Intermolecular Forces — 9701/23/F/M/21/Q4(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]
- 72 Intermolecular Forces — 9701/23/F/M/21/Q5(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]
- 73 Intermolecular Forces — 9701/23/F/M/21/Q6(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]
- 74 Intermolecular Forces — 9701/23/F/M/21/Q7(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]
- 75 Intermolecular Forces — 9701/23/F/M/21/Q8(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]

Question Pack

TOPIC 4 — TOPIC 4: STATES OF MATTER

76 Ideal Gas — 9701/21/M/J/20/Q1(a) Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**77 Ideal Gas — 9701/21/M/J/20/Q2(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**78 Ideal Gas — 9701/21/M/J/20/Q3(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**79 Ideal Gas — 9701/21/M/J/20/Q4(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**80 Ideal Gas — 9701/21/M/J/20/Q5(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**81 Ideal Gas — 9701/21/M/J/20/Q6(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**82 Ideal Gas — 9701/21/M/J/20/Q7(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**83 Ideal Gas — 9701/21/M/J/20/Q8(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**84 Ideal Gas — 9701/21/M/J/20/Q9(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**85 Ideal Gas — 9701/21/M/J/20/Q10(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]

Question Pack

- 86 Real Gases – 9701/22/O/N/22/Q1(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 87 Real Gases – 9701/22/O/N/22/Q2(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 88 Real Gases – 9701/22/O/N/22/Q3(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 89 Real Gases – 9701/22/O/N/22/Q4(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 90 Real Gases – 9701/22/O/N/22/Q5(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 91 Real Gases – 9701/22/O/N/22/Q6(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 92 Real Gases – 9701/22/O/N/22/Q7(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 93 Real Gases – 9701/22/O/N/22/Q8(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 94 Real Gases – 9701/22/O/N/22/Q9(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 95 Real Gases – 9701/22/O/N/22/Q10(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]

Question Pack

TOPIC 5 — TOPIC 5: CHEMICAL ENERGETICS

- 96 Enthalpy — 9701/22/M/J/22/Q1(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 97 Enthalpy — 9701/22/M/J/22/Q2(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 98 Enthalpy — 9701/22/M/J/22/Q3(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 99 Enthalpy — 9701/22/M/J/22/Q4(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 100 Enthalpy — 9701/22/M/J/22/Q5(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 101 Enthalpy — 9701/22/M/J/22/Q6(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 102 Enthalpy — 9701/22/M/J/22/Q7(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 103 Enthalpy — 9701/22/M/J/22/Q8(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 104 Enthalpy — 9701/22/M/J/22/Q9(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 105 Enthalpy — 9701/22/M/J/22/Q10(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 106 Enthalpy — 9701/22/M/J/22/Q11(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 107 Enthalpy — 9701/22/M/J/22/Q12(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 108 Hess Law — 9701/21/O/N/19/Q1(b)** Syllabus 5.2
[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]
(a) Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. [3]

Question Pack

109 Hess Law — 9701/21/O/N/19/Q2(b) Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****110 Hess Law — 9701/21/O/N/19/Q3(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****111 Hess Law — 9701/21/O/N/19/Q4(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****112 Hess Law — 9701/21/O/N/19/Q5(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****113 Hess Law — 9701/21/O/N/19/Q6(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****114 Hess Law — 9701/21/O/N/19/Q7(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****115 Hess Law — 9701/21/O/N/19/Q8(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****116 Hess Law — 9701/21/O/N/19/Q9(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****117 Hess Law — 9701/21/O/N/19/Q10(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]**

Question Pack

118 Hess Law — 9701/21/O/N/19/Q11(b) Syllabus 5.2

[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]

(a)

Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹.

[3]

119 Hess Law — 9701/21/O/N/19/Q12(b) Syllabus 5.2

[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]

(a)

Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹.

[3]

120 Hess Law — 9701/21/O/N/19/Q13(b) Syllabus 5.2

[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]

(a)

Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹.

[3]

Question Pack

TOPIC 6 – TOPIC 6: ELECTROCHEMISTRY

121 Electrolysis – 9701/22/O/N/20/Q2(b) Syllabus 6.2

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

122 Redox Processes – 9701/23/F/M/21/Q1(a) Syllabus 6.1

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

123 Electrolysis – 9701/24/M/J/22/Q2(b) Syllabus 6.2

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

124 Redox Processes – 9701/21/O/N/23/Q1(a) Syllabus 6.1

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

125 Electrolysis – 9701/22/F/M/24/Q2(b) Syllabus 6.2

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

126 Redox Processes – 9701/23/M/J/19/Q1(a) Syllabus 6.1

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

Question Pack

127 Electrolysis — 9701/24/O/N/20/Q2(b) Syllabus 6.2

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

128 Redox Processes — 9701/21/F/M/21/Q1(a) Syllabus 6.1

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

129 Electrolysis — 9701/22/M/J/22/Q2(b) Syllabus 6.2

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 18 A flows for 30 minutes. ($A_r = 59$, charge = +2).

[3]

130 Redox Processes — 9701/23/O/N/23/Q1(a) Syllabus 6.1

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 20 A flows for 30 minutes. ($A_r = 60$, charge = +2).

[3]

131 Electrolysis — 9701/24/F/M/24/Q2(b) Syllabus 6.2

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 22 A flows for 30 minutes. ($A_r = 61$, charge = +2).

[3]

132 Redox Processes — 9701/21/M/J/19/Q1(a) Syllabus 6.1

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 24 A flows for 30 minutes. ($A_r = 62$, charge = +2).

[3]

133 Electrolysis — 9701/22/O/N/20/Q2(b) Syllabus 6.2

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 26 A flows for 30 minutes. ($A_r = 63$, charge = +2).

[3]

134 Redox Processes — 9701/23/F/M/21/Q1(a) Syllabus 6.1

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 28 A flows for 30 minutes. ($A_r = 64$, charge = +2).

[3]

Question Pack

135 Electrolysis – 9701/24/M/J/22/Q2(b) Syllabus 6.2

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 30 A flows for 30 minutes. (Ar = 65, charge = +2).

[3]

136 Redox Processes – 9701/21/O/N/23/Q1(a) Syllabus 6.1

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 32 A flows for 30 minutes. (Ar = 66, charge = +2).

[3]

137 Electrolysis – 9701/22/F/M/24/Q2(b) Syllabus 6.2

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 34 A flows for 30 minutes. (Ar = 67, charge = +2).

[3]

138 Redox Processes – 9701/23/M/J/19/Q1(a) Syllabus 6.1

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 36 A flows for 30 minutes. (Ar = 68, charge = +2).

[3]

139 Electrolysis – 9701/24/O/N/20/Q2(b) Syllabus 6.2

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 38 A flows for 30 minutes. (Ar = 69, charge = +2).

[3]

140 Redox Processes – 9701/21/F/M/21/Q1(a) Syllabus 6.1

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 40 A flows for 30 minutes. (Ar = 70, charge = +2).

[3]

Question Pack

TOPIC 7 — TOPIC 7: EQUILIBRIA

141 Chemical Equilibria — 9701/22/O/N/20/Q3(c) Syllabus 7.2

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ? [1]

A High T, Low P

B Low T, High P

C Low T, Low P

D High T, High P

142 Chemical Equilibria — 9701/23/F/M/21/Q3(c) Syllabus 7.1

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ? [1]

A High T, Low P

B Low T, High P

C Low T, Low P

D High T, High P

143 Chemical Equilibria — 9701/24/M/J/22/Q3(c) Syllabus 7.2

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ? [1]

A High T, Low P

B Low T, High P

C Low T, Low P

D High T, High P

144 Chemical Equilibria — 9701/21/O/N/23/Q3(c) Syllabus 7.1

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ? [1]

A High T, Low P

B Low T, High P

C Low T, Low P

D High T, High P

145 Chemical Equilibria — 9701/22/F/M/24/Q3(c) Syllabus 7.2

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ? [1]

A High T, Low P

B Low T, High P

C Low T, Low P

D High T, High P

146 Chemical Equilibria — 9701/23/M/J/19/Q3(c) Syllabus 7.1

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ? [1]

A High T, Low P

B Low T, High P

C Low T, Low P

D High T, High P

Question Pack

147 Chemical Equilibria — 9701/24/O/N/20/Q3(c) Syllabus 7.2

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ?

- A High T, Low P
- B Low T, High P
- C Low T, Low P
- D High T, High P

[1]**148 Chemical Equilibria — 9701/21/F/M/21/Q3(c)** Syllabus 7.1

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ?

- A High T, Low P
- B Low T, High P
- C Low T, Low P
- D High T, High P

[1]**149 Chemical Equilibria — 9701/22/M/J/22/Q3(c)** Syllabus 7.2

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ?

- A High T, Low P
- B Low T, High P
- C Low T, Low P
- D High T, High P

[1]**150 Chemical Equilibria — 9701/23/O/N/23/Q3(c)** Syllabus 7.1

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ?

- A High T, Low P
- B Low T, High P
- C Low T, Low P
- D High T, High P

[1]**151 Chemical Equilibria — 9701/24/F/M/24/Q3(c)** Syllabus 7.2Consider the dynamic equilibrium: $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$ **(a)**

Define the term dynamic equilibrium.

[2]**(b)**

Explain the effect of adding a catalyst on the position of equilibrium.

[2]**152 Chemical Equilibria — 9701/21/M/J/19/Q3(c)** Syllabus 7.1Consider the dynamic equilibrium: $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$ **(a)**

Define the term dynamic equilibrium.

[2]**(b)**

Explain the effect of adding a catalyst on the position of equilibrium.

[2]

Question Pack

153 Chemical Equilibria — 9701/22/O/N/20/Q3(c) Syllabus 7.2

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

154 Chemical Equilibria — 9701/23/F/M/21/Q3(c) Syllabus 7.1

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

155 Chemical Equilibria — 9701/24/M/J/22/Q3(c) Syllabus 7.2

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

156 Chemical Equilibria — 9701/21/O/N/23/Q3(c) Syllabus 7.1

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

157 Chemical Equilibria — 9701/22/F/M/24/Q3(c) Syllabus 7.2

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

158 Chemical Equilibria — 9701/23/M/J/19/Q3(c) Syllabus 7.1

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

159 Chemical Equilibria — 9701/24/O/N/20/Q3(c) Syllabus 7.2

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

160 Chemical Equilibria — 9701/21/F/M/21/Q3(c) Syllabus 7.1

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

Question Pack

161 Chemical Equilibria – 9701/22/M/J/22/Q3(c) Syllabus 7.2

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

162 Chemical Equilibria – 9701/23/O/N/23/Q3(c) Syllabus 7.1

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

163 Chemical Equilibria – 9701/24/F/M/24/Q3(c) Syllabus 7.2

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

164 Chemical Equilibria – 9701/21/M/J/19/Q3(c) Syllabus 7.1

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

165 Chemical Equilibria – 9701/22/O/N/20/Q3(c) Syllabus 7.2

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

Question Pack

TOPIC 8 — TOPIC 8: REACTION KINETICS

166 Rates of Reaction — 9701/22/O/N/20/Q4(d) Syllabus 8.2

Fig 1.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

167 Rates of Reaction — 9701/23/F/M/21/Q4(d) Syllabus 8.3

Fig 2.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

168 Rates of Reaction — 9701/24/M/J/22/Q4(d) Syllabus 8.1

Fig 3.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

169 Rates of Reaction — 9701/21/O/N/23/Q4(d) Syllabus 8.2

Fig 4.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

170 Rates of Reaction — 9701/22/F/M/24/Q4(d) Syllabus 8.3

Fig 5.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

171 Rates of Reaction — 9701/23/M/J/19/Q4(d) Syllabus 8.1

Fig 6.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

Question Pack

172 Rates of Reaction — 9701/24/O/N/20/Q4(d) Syllabus 8.2

Fig 7.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

173 Rates of Reaction — 9701/21/F/M/21/Q4(d) Syllabus 8.3

Fig 8.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

174 Rates of Reaction — 9701/22/M/J/22/Q4(d) Syllabus 8.1

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

175 Rates of Reaction — 9701/23/O/N/23/Q4(d) Syllabus 8.2

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

176 Rates of Reaction — 9701/24/F/M/24/Q4(d) Syllabus 8.3

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

177 Rates of Reaction — 9701/21/M/J/19/Q4(d) Syllabus 8.1

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

178 Rates of Reaction — 9701/22/O/N/20/Q4(d) Syllabus 8.2

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

179 Rates of Reaction — 9701/23/F/M/21/Q4(d) Syllabus 8.3

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

Question Pack

180 Rates of Reaction — 9701/24/M/J/22/Q4(d) Syllabus 8.1

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

181 Rates of Reaction — 9701/21/O/N/23/Q4(d) Syllabus 8.2

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

182 Rates of Reaction — 9701/22/F/M/24/Q4(d) Syllabus 8.3

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

183 Rates of Reaction — 9701/23/M/J/19/Q4(d) Syllabus 8.1

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

184 Rates of Reaction — 9701/24/O/N/20/Q4(d) Syllabus 8.2

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

185 Rates of Reaction — 9701/21/F/M/21/Q4(d) Syllabus 8.3

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

Question Pack

TOPIC 9 – TOPIC 9: THE PERIODIC TABLE: CHEMICAL PERIODICITY

186 Periodicity – 9701/22/O/N/20/Q5(a) Syllabus 9.2

Fig 1.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

187 Periodicity – 9701/23/F/M/21/Q5(a) Syllabus 9.3

Fig 2.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

188 Periodicity – 9701/24/M/J/22/Q5(a) Syllabus 9.1

Fig 3.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

189 Periodicity – 9701/21/O/N/23/Q5(a) Syllabus 9.2

Fig 4.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

190 Periodicity – 9701/22/F/M/24/Q5(a) Syllabus 9.3

Fig 5.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

191 Periodicity – 9701/23/M/J/19/Q5(a) Syllabus 9.1

Fig 6.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

Question Pack

192 Periodicity — 9701/24/O/N/20/Q5(a) Syllabus 9.2

Fig 7.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

193 Periodicity — 9701/21/F/M/21/Q5(a) Syllabus 9.3

Fig 8.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

194 Periodicity — 9701/22/M/J/22/Q5(a) Syllabus 9.1

Fig 9.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

195 Periodicity — 9701/23/O/N/23/Q5(a) Syllabus 9.2

Fig 10.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

196 Periodicity — 9701/24/F/M/24/Q5(a) Syllabus 9.3

Period 3 elements react with oxygen to form oxides.

- (a) Write an equation for the reaction of sodium with oxygen. [1]
- (b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

197 Periodicity — 9701/21/M/J/19/Q5(a) Syllabus 9.1

Period 3 elements react with oxygen to form oxides.

- (a) Write an equation for the reaction of sodium with oxygen. [1]
- (b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

Question Pack

198 Periodicity – 9701/22/O/N/20/Q5(a) Syllabus 9.2

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

199 Periodicity – 9701/23/F/M/21/Q5(a) Syllabus 9.3

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

200 Periodicity – 9701/24/M/J/22/Q5(a) Syllabus 9.1

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

201 Periodicity – 9701/21/O/N/23/Q5(a) Syllabus 9.2

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

202 Periodicity – 9701/22/F/M/24/Q5(a) Syllabus 9.3

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

203 Periodicity – 9701/23/M/J/19/Q5(a) Syllabus 9.1

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

204 Periodicity – 9701/24/O/N/20/Q5(a) Syllabus 9.2

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

205 Periodicity – 9701/21/F/M/21/Q5(a) Syllabus 9.3

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

Question Pack

206 Periodicity – 9701/22/M/J/22/Q5(a) Syllabus 9.1

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

207 Periodicity – 9701/23/O/N/23/Q5(a) Syllabus 9.2

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

208 Periodicity – 9701/24/F/M/24/Q5(a) Syllabus 9.3

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

209 Periodicity – 9701/21/M/J/19/Q5(a) Syllabus 9.1

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

210 Periodicity – 9701/22/O/N/20/Q5(a) Syllabus 9.2

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

Question Pack

TOPIC 10 — TOPIC 10.1: GROUP 2

211 GROUP 2 — 9701/21/O/N/22/Q8(c) Syllabus 10.1

""

- (a) State the flame colour of barium. [1]

212 GROUP 2 — 9701/24/F/M/22/Q6(b) Syllabus 10.1

""

- (a) Explain the trend in thermal stability of Group 2 carbonates. [3]

213 GROUP 2 — 9701/22/F/M/22/Q7(a) Syllabus 10.1

""

- (a) Write an equation for the thermal decomposition of magnesium nitrate. [2]

214 GROUP 2 — 9701/21/M/J/19/Q7(b) Syllabus 10.1

""

- (a) Compare the reactivity of magnesium and barium with water. [2]

215 GROUP 2 — 9701/21/F/M/25/Q1(c) Syllabus 10.1

"[DIAGRAM: Experimental setup showing thermal decomposition of Group 2 carbonates]"

- (a) Describe the trend in solubility of Group 2 hydroxides down the group. [1]

216 GROUP 2 — 9701/23/F/M/24/Q4(b) Syllabus 10.1

""

- (a) State the flame colour of barium. [1]

217 GROUP 2 — 9701/22/M/J/25/Q8(b) Syllabus 10.1

""

- (a) Explain the trend in thermal stability of Group 2 carbonates. [3]

218 GROUP 2 — 9701/24/F/M/25/Q4(a) Syllabus 10.1

""

- (a) Write an equation for the thermal decomposition of magnesium nitrate. [2]

219 GROUP 2 — 9701/22/M/J/21/Q8(c) Syllabus 10.1

""

- (a) Compare the reactivity of magnesium and barium with water. [2]

220 GROUP 2 — 9701/21/O/N/19/Q4(b) Syllabus 10.1

"[DIAGRAM: Experimental setup showing thermal decomposition of Group 2 carbonates]"

- (a) Describe the trend in solubility of Group 2 hydroxides down the group. [1]

Question Pack

221 GROUP 2 — 9701/24/O/N/20/Q2(c) Syllabus 10.1

....

- (a) State the flame colour of barium. [1]

222 GROUP 2 — 9701/22/F/M/20/Q8(c) Syllabus 10.1

....

- (a) Explain the trend in thermal stability of Group 2 carbonates. [3]

223 GROUP 2 — 9701/24/F/M/25/Q3(b) Syllabus 10.1

....

- (a) Write an equation for the thermal decomposition of magnesium nitrate. [2]

224 GROUP 2 — 9701/22/O/N/24/Q5(c) Syllabus 10.1

....

- (a) Compare the reactivity of magnesium and barium with water. [2]

225 GROUP 2 — 9701/22/M/J/19/Q6(c) Syllabus 10.1

"[DIAGRAM: Experimental setup showing thermal decomposition of Group 2 carbonates]"

- (a) Describe the trend in solubility of Group 2 hydroxides down the group. [1]

226 GROUP 2 — 9701/21/M/J/23/Q8(a) Syllabus 10.1

....

- (a) State the flame colour of barium. [1]

227 GROUP 2 — 9701/23/F/M/21/Q7(a) Syllabus 10.1

....

- (a) Explain the trend in thermal stability of Group 2 carbonates. [3]

228 GROUP 2 — 9701/22/O/N/20/Q2(b) Syllabus 10.1

....

- (a) Write an equation for the thermal decomposition of magnesium nitrate. [2]

229 GROUP 2 — 9701/24/M/J/23/Q9(c) Syllabus 10.1

....

- (a) Compare the reactivity of magnesium and barium with water. [2]

230 GROUP 2 — 9701/23/O/N/20/Q10(c) Syllabus 10.1

"[DIAGRAM: Experimental setup showing thermal decomposition of Group 2 carbonates]"

- (a) Describe the trend in solubility of Group 2 hydroxides down the group. [1]

Question Pack

TOPIC 11 — TOPIC 11.1: GROUP 17

231 GROUP 17 — 9701/21/F/M/19/Q3(b) Syllabus 11.2

- (a) Explain why silver chloride dissolves in dilute aqueous ammonia but silver iodide does not. [2]

232 GROUP 17 — 9701/24/M/J/25/Q9(a) Syllabus 11.1

- (a) Write an equation for the reaction of chlorine with cold aqueous sodium hydroxide. [2]

233 GROUP 17 — 9701/22/F/M/20/Q2(c) Syllabus 11.2

- (a) Suggest why adding aqueous silver nitrate followed by aqueous ammonia is a good test for halide ions. [2]

234 GROUP 17 — 9701/22/O/N/21/Q10(a) Syllabus 11.2

- (a) Explain the use of chlorine in water treatment. [2]

235 GROUP 17 — 9701/21/O/N/21/Q5(a) Syllabus 11.2

- (a) Describe the trend in boiling points of the halogens. [2]

236 GROUP 17 — 9701/24/F/M/24/Q10(c) Syllabus 11.1

"Fig. 2.1 shows the trend in boiling points of the halogens."

- (a) Explain why silver chloride dissolves in dilute aqueous ammonia but silver iodide does not. [2]

237 GROUP 17 — 9701/21/F/M/22/Q5(a) Syllabus 11.1

- (a) Write an equation for the reaction of chlorine with cold aqueous sodium hydroxide. [2]

238 GROUP 17 — 9701/22/O/N/20/Q4(b) Syllabus 11.3

- (a) Suggest why adding aqueous silver nitrate followed by aqueous ammonia is a good test for halide ions. [2]

239 GROUP 17 — 9701/21/M/J/19/Q2(c) Syllabus 11.3

- (a) Explain the use of chlorine in water treatment. [2]

240 GROUP 17 — 9701/21/O/N/25/Q5(b) Syllabus 11.3

- (a) Describe the trend in boiling points of the halogens. [2]

Question Pack

241 GROUP 17 – 9701/24/F/M/21/Q5(b) Syllabus 11.2

....

- (a) Explain why silver chloride dissolves in dilute aqueous ammonia but silver iodide does not. [2]

242 GROUP 17 – 9701/22/M/J/24/Q9(a) Syllabus 11.1

"Fig. 2.1 shows the trend in boiling points of the halogens."

- (a) Write an equation for the reaction of chlorine with cold aqueous sodium hydroxide. [2]

243 GROUP 17 – 9701/21/M/J/19/Q4(c) Syllabus 11.2

....

- (a) Suggest why adding aqueous silver nitrate followed by aqueous ammonia is a good test for halide ions. [2]

244 GROUP 17 – 9701/24/F/M/20/Q1(c) Syllabus 11.3

....

- (a) Explain the use of chlorine in water treatment. [2]

245 GROUP 17 – 9701/24/O/N/23/Q8(b) Syllabus 11.3

....

- (a) Describe the trend in boiling points of the halogens. [2]

246 GROUP 17 – 9701/24/M/J/20/Q8(b) Syllabus 11.2

....

- (a) Explain why silver chloride dissolves in dilute aqueous ammonia but silver iodide does not. [2]

247 GROUP 17 – 9701/23/F/M/21/Q4(a) Syllabus 11.1

....

- (a) Write an equation for the reaction of chlorine with cold aqueous sodium hydroxide. [2]

248 GROUP 17 – 9701/22/F/M/25/Q8(a) Syllabus 11.3

"Fig. 2.1 shows the trend in boiling points of the halogens."

- (a) Suggest why adding aqueous silver nitrate followed by aqueous ammonia is a good test for halide ions. [2]

249 GROUP 17 – 9701/23/M/J/19/Q4(a) Syllabus 11.3

....

- (a) Explain the use of chlorine in water treatment. [2]

250 GROUP 17 – 9701/21/M/J/21/Q9(c) Syllabus 11.2

....

- (a) Describe the trend in boiling points of the halogens. [2]

Question Pack

251 GROUP 17 – 9701/22/F/M/21/Q6(b) Syllabus 11.1

- (a) Explain why silver chloride dissolves in dilute aqueous ammonia but silver iodide does not. [2]

252 GROUP 17 – 9701/21/O/N/20/Q5(a) Syllabus 11.1

- (a) Write an equation for the reaction of chlorine with cold aqueous sodium hydroxide. [2]

253 GROUP 17 – 9701/22/M/J/23/Q6(b) Syllabus 11.2

- (a) Suggest why adding aqueous silver nitrate followed by aqueous ammonia is a good test for halide ions. [2]

254 GROUP 17 – 9701/22/M/J/23/Q10(a) Syllabus 11.4

"Fig. 2.1 shows the trend in boiling points of the halogens."

- (a) Explain the use of chlorine in water treatment. [2]

255 GROUP 17 – 9701/21/F/M/22/Q4(c) Syllabus 11.4

- (a) Describe the trend in boiling points of the halogens. [2]

Question Pack

TOPIC 12 — TOPIC 12.1: NITROGEN AND SULFUR

256 NITROGEN AND SULFUR — 9701/23/M/J/22/Q6(c) Syllabus 12.1

""

(a) Describe the environmental consequences of sulfur dioxide emissions. [2]

257 NITROGEN AND SULFUR — 9701/21/F/M/25/Q1(a) Syllabus 12.1

""

(a) Write the equations for the formation of acid rain from nitrogen dioxide. [1]

258 NITROGEN AND SULFUR — 9701/24/F/M/21/Q9(c) Syllabus 12.1

""

(a) Explain how a catalytic converter removes NO from exhaust gases. [2]

259 NITROGEN AND SULFUR — 9701/22/F/M/20/Q8(b) Syllabus 12.1

""

(a) State a natural source of sulfur dioxide in the atmosphere. [1]

260 NITROGEN AND SULFUR — 9701/24/M/J/24/Q6(b) Syllabus 12.1

"[DIAGRAM: Formation of acid rain from atmospheric pollutants]"

(a) Explain how nitrogen oxides are formed in a car engine. [2]

261 NITROGEN AND SULFUR — 9701/21/O/N/22/Q6(b) Syllabus 12.1

""

(a) Describe the environmental consequences of sulfur dioxide emissions. [2]

262 NITROGEN AND SULFUR — 9701/22/F/M/25/Q3(b) Syllabus 12.1

""

(a) Write the equations for the formation of acid rain from nitrogen dioxide. [1]

263 NITROGEN AND SULFUR — 9701/21/F/M/21/Q1(b) Syllabus 12.1

""

(a) Explain how a catalytic converter removes NO from exhaust gases. [2]

264 NITROGEN AND SULFUR — 9701/24/F/M/23/Q3(c) Syllabus 12.1

""

(a) State a natural source of sulfur dioxide in the atmosphere. [1]

265 NITROGEN AND SULFUR — 9701/23/F/M/23/Q2(b) Syllabus 12.1

"[DIAGRAM: Formation of acid rain from atmospheric pollutants]"

(a) Explain how nitrogen oxides are formed in a car engine. [2]

Question Pack

266 NITROGEN AND SULFUR – 9701/22/M/J/20/Q2(a) Syllabus 12.1

....

- (a) Describe the environmental consequences of sulfur dioxide emissions. [2]

267 NITROGEN AND SULFUR – 9701/21/F/M/24/Q5(c) Syllabus 12.1

....

- (a) Write the equations for the formation of acid rain from nitrogen dioxide. [1]

268 NITROGEN AND SULFUR – 9701/23/F/M/21/Q10(b) Syllabus 12.1

....

- (a) Explain how a catalytic converter removes NO from exhaust gases. [2]

269 NITROGEN AND SULFUR – 9701/24/F/M/25/Q9(b) Syllabus 12.1

....

- (a) State a natural source of sulfur dioxide in the atmosphere. [1]

270 NITROGEN AND SULFUR – 9701/23/F/M/23/Q7(b) Syllabus 12.1

"[DIAGRAM: Formation of acid rain from atmospheric pollutants]"

- (a) Explain how nitrogen oxides are formed in a car engine. [2]

271 NITROGEN AND SULFUR – 9701/22/O/N/20/Q3(a) Syllabus 12.1

....

- (a) Describe the environmental consequences of sulfur dioxide emissions. [2]

272 NITROGEN AND SULFUR – 9701/21/O/N/23/Q6(b) Syllabus 12.1

....

- (a) Write the equations for the formation of acid rain from nitrogen dioxide. [1]

273 NITROGEN AND SULFUR – 9701/22/F/M/20/Q6(c) Syllabus 12.1

....

- (a) Explain how a catalytic converter removes NO from exhaust gases. [2]

274 NITROGEN AND SULFUR – 9701/22/F/M/25/Q2(a) Syllabus 12.1

....

- (a) State a natural source of sulfur dioxide in the atmosphere. [1]

275 NITROGEN AND SULFUR – 9701/22/M/J/22/Q6(a) Syllabus 12.1

"[DIAGRAM: Formation of acid rain from atmospheric pollutants]"

- (a) Explain how nitrogen oxides are formed in a car engine. [2]

Question Pack

TOPIC 13 — TOPIC 13.1: AN INTRODUCTION TO AS LEVEL ORGANIC CHEMISTRY

276 ORGANIC CHEMISTRY — 9701/21/M/J/23/Q8(c) Syllabus 13.3

...

- (a) Explain the difference between structural isomerism and stereoisomerism. [2]

277 ORGANIC CHEMISTRY — 9701/21/F/M/21/Q3(b) Syllabus 13.1

...

- (a) State the Cahn-Ingold-Prelog priority rules for E/Z isomerism. [2]

278 ORGANIC CHEMISTRY — 9701/23/O/N/24/Q4(c) Syllabus 13.4

...

- (a) Identify the functional groups in an amino acid. [2]

279 ORGANIC CHEMISTRY — 9701/23/M/J/21/Q2(a) Syllabus 13.4

"Fig. 3.2 shows the skeletal formula of an organic molecule."

- (a) Describe the oxidation of a primary alcohol. [3]

280 ORGANIC CHEMISTRY — 9701/22/M/J/24/Q6(b) Syllabus 13.1

...

- (a) Give the IUPAC name for $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$. [1]

281 ORGANIC CHEMISTRY — 9701/22/M/J/19/Q7(a) Syllabus 13.4

...

- (a) Explain the difference between structural isomerism and stereoisomerism. [2]

282 ORGANIC CHEMISTRY — 9701/23/M/J/24/Q5(a) Syllabus 13.3

...

- (a) State the Cahn-Ingold-Prelog priority rules for E/Z isomerism. [2]

283 ORGANIC CHEMISTRY — 9701/22/O/N/25/Q5(a) Syllabus 13.1

"Fig. 3.2 shows the skeletal formula of an organic molecule."

- (a) Identify the functional groups in an amino acid. [2]

284 ORGANIC CHEMISTRY — 9701/21/M/J/24/Q10(c) Syllabus 13.1

...

- (a) Describe the oxidation of a primary alcohol. [3]

285 ORGANIC CHEMISTRY — 9701/21/O/N/22/Q2(b) Syllabus 13.4

...

- (a) Give the IUPAC name for $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$. [1]

Question Pack

286 ORGANIC CHEMISTRY – 9701/23/F/M/20/Q10(c) Syllabus 13.1

- (a) Explain the difference between structural isomerism and stereoisomerism. [2]

287 ORGANIC CHEMISTRY – 9701/21/F/M/20/Q1(a) Syllabus 13.3

"Fig. 3.2 shows the skeletal formula of an organic molecule."

- (a) State the Cahn-Ingold-Prelog priority rules for E/Z isomerism. [2]

288 ORGANIC CHEMISTRY – 9701/21/O/N/23/Q7(c) Syllabus 13.4

- (a) Identify the functional groups in an amino acid. [2]

289 ORGANIC CHEMISTRY – 9701/22/F/M/22/Q1(b) Syllabus 13.1

- (a) Describe the oxidation of a primary alcohol. [3]

290 ORGANIC CHEMISTRY – 9701/23/F/M/22/Q2(c) Syllabus 13.1

- (a) Give the IUPAC name for $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$. [1]

291 ORGANIC CHEMISTRY – 9701/21/M/J/19/Q3(b) Syllabus 13.2

"Fig. 3.2 shows the skeletal formula of an organic molecule."

- (a) Explain the difference between structural isomerism and stereoisomerism. [2]

292 ORGANIC CHEMISTRY – 9701/23/O/N/25/Q5(b) Syllabus 13.4

- (a) State the Cahn-Ingold-Prelog priority rules for E/Z isomerism. [2]

293 ORGANIC CHEMISTRY – 9701/22/M/J/21/Q6(b) Syllabus 13.2

- (a) Identify the functional groups in an amino acid. [2]

294 ORGANIC CHEMISTRY – 9701/21/F/M/20/Q4(c) Syllabus 13.1

- (a) Describe the oxidation of a primary alcohol. [3]

295 ORGANIC CHEMISTRY – 9701/24/F/M/22/Q4(c) Syllabus 13.1

"Fig. 3.2 shows the skeletal formula of an organic molecule."

- (a) Give the IUPAC name for $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$. [1]

Question Pack

296 ORGANIC CHEMISTRY – 9701/23/F/M/23/Q8(c) Syllabus 13.1

- (a) Explain the difference between structural isomerism and stereoisomerism. [2]

297 ORGANIC CHEMISTRY – 9701/22/M/J/24/Q10(b) Syllabus 13.1

- (a) State the Cahn-Ingold-Prelog priority rules for E/Z isomerism. [2]

298 ORGANIC CHEMISTRY – 9701/23/M/J/22/Q10(c) Syllabus 13.4

- (a) Identify the functional groups in an amino acid. [2]

299 ORGANIC CHEMISTRY – 9701/22/O/N/22/Q5(c) Syllabus 13.2

"Fig. 3.2 shows the skeletal formula of an organic molecule."

- (a) Describe the oxidation of a primary alcohol. [3]

300 ORGANIC CHEMISTRY – 9701/22/F/M/22/Q10(c) Syllabus 13.1

- (a) Give the IUPAC name for $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$. [1]

Question Pack

TOPIC 14 — TOPIC 14: HYDROCARBONS

301 Alkanes — 9701/21/M/J/2019/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing ethane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

302 Alkanes — 9701/22/O/N/2020/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing propane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

303 Alkanes — 9701/23/F/M/2021/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing butane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

304 Alkanes — 9701/21/M/J/2022/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing pentane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

305 Alkanes — 9701/22/O/N/2023/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing hexane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

306 Alkanes — 9701/23/F/M/2024/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing ethane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

307 Alkanes — 9701/21/M/J/2019/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing propane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

Question Pack

308 Alkanes — 9701/22/O/N/2020/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing butane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

309 Alkanes — 9701/23/F/M/2021/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing pentane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

310 Alkanes — 9701/21/M/J/2022/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing hexane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

311 Alkanes — 9701/22/O/N/2023/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing ethane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

312 Alkanes — 9701/23/F/M/2024/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing propane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

313 Alkanes — 9701/21/M/J/2019/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing butane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

314 Alkenes — 9701/21/M/J/2020/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of ethene.

(a) Describe the test for unsaturation and state the observation when ethene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to ethene. Include all relevant curly arrows, charges, and dipoles. [4]

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315 Alkenes — 9701/22/O/N/2021/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of propene.

(a) Describe the test for unsaturation and state the observation when propene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to propene. Include all relevant curly arrows, charges, and dipoles. [4]

316 Alkenes — 9701/23/F/M/2022/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of but-1-ene.

(a) Describe the test for unsaturation and state the observation when but-1-ene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to but-1-ene. Include all relevant curly arrows, charges, and dipoles. [4]

317 Alkenes — 9701/21/M/J/2023/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of but-2-ene.

(a) Describe the test for unsaturation and state the observation when but-2-ene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to but-2-ene. Include all relevant curly arrows, charges, and dipoles. [4]

318 Alkenes — 9701/22/O/N/2024/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of ethene.

(a) Describe the test for unsaturation and state the observation when ethene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to ethene. Include all relevant curly arrows, charges, and dipoles. [4]

319 Alkenes — 9701/23/F/M/2020/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of propene.

(a) Describe the test for unsaturation and state the observation when propene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to propene. Include all relevant curly arrows, charges, and dipoles. [4]

320 Alkenes — 9701/21/M/J/2021/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of but-1-ene.

(a) Describe the test for unsaturation and state the observation when but-1-ene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to but-1-ene. Include all relevant curly arrows, charges, and dipoles. [4]

321 Alkenes — 9701/22/O/N/2022/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of but-2-ene.

(a) Describe the test for unsaturation and state the observation when but-2-ene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to but-2-ene. Include all relevant curly arrows, charges, and dipoles. [4]

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322 Alkenes – 9701/23/F/M/2023/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of ethene.

(a) Describe the test for unsaturation and state the observation when ethene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to ethene. Include all relevant curly arrows, charges, and dipoles. [4]

323 Alkenes – 9701/21/M/J/2024/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of propene.

(a) Describe the test for unsaturation and state the observation when propene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to propene. Include all relevant curly arrows, charges, and dipoles. [4]

324 Alkenes – 9701/22/O/N/2020/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of but-1-ene.

(a) Describe the test for unsaturation and state the observation when but-1-ene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to but-1-ene. Include all relevant curly arrows, charges, and dipoles. [4]

325 Alkenes – 9701/23/F/M/2021/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of but-2-ene.

(a) Describe the test for unsaturation and state the observation when but-2-ene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to but-2-ene. Include all relevant curly arrows, charges, and dipoles. [4]

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TOPIC 15 — TOPIC 15: HALOGEN COMPOUNDS

326 Halogenoalkanes — 9701/21/M/J/2019/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-chloropropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

327 Halogenoalkanes — 9701/22/O/N/2020/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-bromobutane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

328 Halogenoalkanes — 9701/23/F/M/2021/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-chloro-2-methylpropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

329 Halogenoalkanes — 9701/21/M/J/2022/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-iodopentane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

330 Halogenoalkanes — 9701/22/O/N/2023/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-chloropropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

331 Halogenoalkanes — 9701/23/F/M/2024/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-bromobutane. [2]

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- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

332 Halogenoalkanes – 9701/21/M/J/2019/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-chloro-2-methylpropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

333 Halogenoalkanes – 9701/22/O/N/2020/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-iodopentane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

334 Halogenoalkanes – 9701/23/F/M/2021/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-chloropropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

335 Halogenoalkanes – 9701/21/M/J/2022/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-bromobutane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

336 Halogenoalkanes – 9701/22/O/N/2023/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-chloro-2-methylpropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

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337 Halogenoalkanes – 9701/23/F/M/2024/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-iodopentane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

338 Halogenoalkanes – 9701/21/M/J/2019/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-chloropropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

339 Halogenoalkanes – 9701/22/O/N/2020/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-bromobutane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

340 Halogenoalkanes – 9701/23/F/M/2021/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-chloro-2-methylpropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

341 Halogenoalkanes – 9701/21/M/J/2022/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-iodopentane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

342 Halogenoalkanes – 9701/22/O/N/2023/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-chloropropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]

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- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

343 Halogenoalkanes – 9701/23/F/M/2024/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-bromobutane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

344 Halogenoalkanes – 9701/21/M/J/2019/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-chloro-2-methylpropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

345 Halogenoalkanes – 9701/22/O/N/2020/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-iodopentane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

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TOPIC 16 — TOPIC 16: HYDROXY COMPOUNDS

346 Alcohols — 9701/21/M/J/2019/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of ethanol]

- (a) Classify ethanol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when ethanol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of ethanol. [1]

347 Alcohols — 9701/22/O/N/2020/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of propan-2-ol]

- (a) Classify propan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when propan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of propan-2-ol. [1]

348 Alcohols — 9701/23/F/M/2021/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of 2-methylpropan-2-ol]

- (a) Classify 2-methylpropan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when 2-methylpropan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of 2-methylpropan-2-ol. [1]

349 Alcohols — 9701/21/M/J/2022/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of butan-1-ol]

- (a) Classify butan-1-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when butan-1-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of butan-1-ol. [1]

350 Alcohols — 9701/22/O/N/2023/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of ethanol]

- (a) Classify ethanol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when ethanol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of ethanol. [1]

351 Alcohols — 9701/23/F/M/2024/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of propan-2-ol]

- (a) Classify propan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when propan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]

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- (c) State the conditions required for the dehydration of propan-2-ol. [1]

352 Alcohols — 9701/21/M/J/2019/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of 2-methylpropan-2-ol]

- (a) Classify 2-methylpropan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when 2-methylpropan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of 2-methylpropan-2-ol. [1]

353 Alcohols — 9701/22/O/N/2020/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of butan-1-ol]

- (a) Classify butan-1-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when butan-1-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of butan-1-ol. [1]

354 Alcohols — 9701/23/F/M/2021/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of ethanol]

- (a) Classify ethanol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when ethanol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of ethanol. [1]

355 Alcohols — 9701/21/M/J/2022/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of propan-2-ol]

- (a) Classify propan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when propan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of propan-2-ol. [1]

356 Alcohols — 9701/22/O/N/2023/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of 2-methylpropan-2-ol]

- (a) Classify 2-methylpropan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when 2-methylpropan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of 2-methylpropan-2-ol. [1]

357 Alcohols — 9701/23/F/M/2024/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of butan-1-ol]

- (a) Classify butan-1-ol as primary, secondary, or tertiary. Explain your answer. [2]

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- (b) Describe what is observed when butan-1-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of butan-1-ol. [1]

358 Alcohols – 9701/21/M/J/2019/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of ethanol]

- (a) Classify ethanol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when ethanol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of ethanol. [1]

359 Alcohols – 9701/22/O/N/2020/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of propan-2-ol]

- (a) Classify propan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when propan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of propan-2-ol. [1]

360 Alcohols – 9701/23/F/M/2021/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of 2-methylpropan-2-ol]

- (a) Classify 2-methylpropan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when 2-methylpropan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of 2-methylpropan-2-ol. [1]

361 Alcohols – 9701/21/M/J/2022/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of butan-1-ol]

- (a) Classify butan-1-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when butan-1-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of butan-1-ol. [1]

362 Alcohols – 9701/22/O/N/2023/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of ethanol]

- (a) Classify ethanol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when ethanol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of ethanol. [1]

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363 Alcohols – 9701/23/F/M/2024/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of propan-2-ol]

- (a) Classify propan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when propan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of propan-2-ol. [1]

364 Alcohols – 9701/21/M/J/2019/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of 2-methylpropan-2-ol]

- (a) Classify 2-methylpropan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when 2-methylpropan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of 2-methylpropan-2-ol. [1]

365 Alcohols – 9701/22/O/N/2020/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of butan-1-ol]

- (a) Classify butan-1-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when butan-1-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of butan-1-ol. [1]

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TOPIC 17 — TOPIC 17: CARBONYL COMPOUNDS

366 Carbonyls — 9701/21/M/J/2019/Q5(d) Syllabus 17.1

Compound X is ethanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to ethanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for ethanal and state the name of the organic product. [2]

367 Carbonyls — 9701/22/O/N/2020/Q5(d) Syllabus 17.1

Compound X is propanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to propanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for propanone and state the name of the organic product. [2]

368 Carbonyls — 9701/23/F/M/2021/Q5(d) Syllabus 17.1

Compound X is butanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanal and state the name of the organic product. [2]

369 Carbonyls — 9701/21/M/J/2022/Q5(d) Syllabus 17.1

Compound X is butanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanone and state the name of the organic product. [2]

370 Carbonyls — 9701/22/O/N/2023/Q5(d) Syllabus 17.1

Compound X is ethanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to ethanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for ethanal and state the name of the organic product. [2]

371 Carbonyls — 9701/23/F/M/2024/Q5(d) Syllabus 17.1

Compound X is propanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]

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- (b) Outline the mechanism for the nucleophilic addition of HCN to propanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for propanone and state the name of the organic product. [2]

372 Carbonyls — 9701/21/M/J/2019/Q5(d) Syllabus 17.1

Compound X is butanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanal and state the name of the organic product. [2]

373 Carbonyls — 9701/22/O/N/2020/Q5(d) Syllabus 17.1

Compound X is butanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanone and state the name of the organic product. [2]

374 Carbonyls — 9701/23/F/M/2021/Q5(d) Syllabus 17.1

Compound X is ethanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to ethanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for ethanal and state the name of the organic product. [2]

375 Carbonyls — 9701/21/M/J/2022/Q5(d) Syllabus 17.1

Compound X is propanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to propanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for propanone and state the name of the organic product. [2]

376 Carbonyls — 9701/22/O/N/2023/Q5(d) Syllabus 17.1

Compound X is butanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanal and state the name of the organic product. [2]

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377 Carbonyls — 9701/23/F/M/2024/Q5(d) Syllabus 17.1

Compound X is butanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanone and state the name of the organic product. [2]

378 Carbonyls — 9701/21/M/J/2019/Q5(d) Syllabus 17.1

Compound X is ethanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to ethanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for ethanal and state the name of the organic product. [2]

379 Carbonyls — 9701/22/O/N/2020/Q5(d) Syllabus 17.1

Compound X is propanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to propanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for propanone and state the name of the organic product. [2]

380 Carbonyls — 9701/23/F/M/2021/Q5(d) Syllabus 17.1

Compound X is butanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanal and state the name of the organic product. [2]

381 Carbonyls — 9701/21/M/J/2022/Q5(d) Syllabus 17.1

Compound X is butanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanone and state the name of the organic product. [2]

382 Carbonyls — 9701/22/O/N/2023/Q5(d) Syllabus 17.1

Compound X is ethanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]

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- (b) Outline the mechanism for the nucleophilic addition of HCN to ethanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for ethanal and state the name of the organic product. [2]

383 Carbonyls – 9701/23/F/M/2024/Q5(d) Syllabus 17.1

Compound X is propanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to propanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for propanone and state the name of the organic product. [2]

384 Carbonyls – 9701/21/M/J/2019/Q5(d) Syllabus 17.1

Compound X is butanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanal and state the name of the organic product. [2]

385 Carbonyls – 9701/22/O/N/2020/Q5(d) Syllabus 17.1

Compound X is butanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanone and state the name of the organic product. [2]

Question Pack

TOPIC 18 — TOPIC 18: CARBOXYLIC ACIDS AND DERIVATIVES

386 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q1(a) Syllabus 18.1

Question 1 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

387 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q2(a) Syllabus 18.1

Question 2 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH₄
 - B LiAlH₄
 - C H₂/Ni
 - D Sn/HCl

388 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q3(a) Syllabus 18.1

Question 3 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

389 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q4(a) Syllabus 18.1

Question 4 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH₄
 - B LiAlH₄
 - C H₂/Ni
 - D Sn/HCl

390 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q5(a) Syllabus 18.1

Question 5 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

391 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q6(a) Syllabus 18.1

Question 6 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH₄
 - B LiAlH₄
 - C H₂/Ni
 - D Sn/HCl

392 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q7(a) Syllabus 18.1

Question 7 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

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393 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q8(a) Syllabus 18.1

Question 8 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH_4
B LiAlH_4
C H_2/Ni
D Sn/HCl

394 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q9(a) Syllabus 18.1

Question 9 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

395 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q10(a) Syllabus 18.1

Question 10 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH_4
B LiAlH_4
C H_2/Ni
D Sn/HCl

396 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q11(a) Syllabus 18.2

Question 11 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

397 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q12(a) Syllabus 18.2

Question 12 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH_4
B LiAlH_4
C H_2/Ni
D Sn/HCl

398 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q13(a) Syllabus 18.2

Question 13 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

399 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q14(a) Syllabus 18.2

Question 14 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH_4
B LiAlH_4
C H_2/Ni
D Sn/HCl

Question Pack

400 Carboxylic Acids and Derivatives – 9701/22/M/J/23/Q15(a) Syllabus 18.2

Question 15 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

401 Carboxylic Acids and Derivatives – 9701/22/M/J/23/Q16(a) Syllabus 18.2

Question 16 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH₄
 - B LiAlH₄
 - C H₂/Ni
 - D Sn/HCl

402 Carboxylic Acids and Derivatives – 9701/22/M/J/23/Q17(a) Syllabus 18.2

Question 17 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

403 Carboxylic Acids and Derivatives – 9701/22/M/J/23/Q18(a) Syllabus 18.2

Question 18 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH₄
 - B LiAlH₄
 - C H₂/Ni
 - D Sn/HCl

404 Carboxylic Acids and Derivatives – 9701/22/M/J/23/Q19(a) Syllabus 18.2

Question 19 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

405 Carboxylic Acids and Derivatives – 9701/22/M/J/23/Q20(a) Syllabus 18.2

Question 20 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH₄
 - B LiAlH₄
 - C H₂/Ni
 - D Sn/HCl

Question Pack

TOPIC 19 — TOPIC 19: NITROGEN COMPOUNDS

406 Nitrogen Compounds — 9701/21/O/N/23/Q1(b) Syllabus 19.1

Question 1 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

407 Nitrogen Compounds — 9701/21/O/N/23/Q2(b) Syllabus 19.1

Question 2 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

408 Nitrogen Compounds — 9701/21/O/N/23/Q3(b) Syllabus 19.1

Question 3 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

409 Nitrogen Compounds — 9701/21/O/N/23/Q4(b) Syllabus 19.1

Question 4 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

410 Nitrogen Compounds — 9701/21/O/N/23/Q5(b) Syllabus 19.1

Question 5 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

411 Nitrogen Compounds — 9701/21/O/N/23/Q6(b) Syllabus 19.1

Question 6 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

412 Nitrogen Compounds — 9701/21/O/N/23/Q7(b) Syllabus 19.1

Question 7 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

413 Nitrogen Compounds — 9701/21/O/N/23/Q8(b) Syllabus 19.1

Question 8 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

414 Nitrogen Compounds — 9701/21/O/N/23/Q9(b) Syllabus 19.1

Question 9 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

415 Nitrogen Compounds — 9701/21/O/N/23/Q10(b) Syllabus 19.1

Question 10 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

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416 Nitrogen Compounds — 9701/21/O/N/23/Q11(b) Syllabus 19.2

Question 11 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

417 Nitrogen Compounds — 9701/21/O/N/23/Q12(b) Syllabus 19.2

Question 12 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

418 Nitrogen Compounds — 9701/21/O/N/23/Q13(b) Syllabus 19.2

Question 13 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

419 Nitrogen Compounds — 9701/21/O/N/23/Q14(b) Syllabus 19.2

Question 14 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

420 Nitrogen Compounds — 9701/21/O/N/23/Q15(b) Syllabus 19.2

Question 15 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

421 Nitrogen Compounds — 9701/21/O/N/23/Q16(b) Syllabus 19.2

Question 16 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

422 Nitrogen Compounds — 9701/21/O/N/23/Q17(b) Syllabus 19.2

Question 17 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

423 Nitrogen Compounds — 9701/21/O/N/23/Q18(b) Syllabus 19.2

Question 18 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

424 Nitrogen Compounds — 9701/21/O/N/23/Q19(b) Syllabus 19.2

Question 19 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

425 Nitrogen Compounds — 9701/21/O/N/23/Q20(b) Syllabus 19.2

Question 20 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

Question Pack

TOPIC 20 – TOPIC 20: POLYMERISATION

426 Polymerisation – 9701/23/F/M/24/Q1(c) Syllabus 20.1

Question 1 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

427 Polymerisation – 9701/23/F/M/24/Q2(c) Syllabus 20.1

Question 2 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

428 Polymerisation – 9701/23/F/M/24/Q3(c) Syllabus 20.1

Question 3 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

429 Polymerisation – 9701/23/F/M/24/Q4(c) Syllabus 20.1

Question 4 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

430 Polymerisation – 9701/23/F/M/24/Q5(c) Syllabus 20.1

Question 5 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

431 Polymerisation – 9701/23/F/M/24/Q6(c) Syllabus 20.1

Question 6 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

432 Polymerisation – 9701/23/F/M/24/Q7(c) Syllabus 20.1

Question 7 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

433 Polymerisation – 9701/23/F/M/24/Q8(c) Syllabus 20.1

Question 8 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

434 Polymerisation – 9701/23/F/M/24/Q9(c) Syllabus 20.1

Question 9 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

435 Polymerisation – 9701/23/F/M/24/Q10(c) Syllabus 20.1

Question 10 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

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436 Polymerisation — 9701/23/F/M/24/Q11(c) Syllabus 20.1

Question 11 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

437 Polymerisation — 9701/23/F/M/24/Q12(c) Syllabus 20.1

Question 12 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

438 Polymerisation — 9701/23/F/M/24/Q13(c) Syllabus 20.1

Question 13 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

439 Polymerisation — 9701/23/F/M/24/Q14(c) Syllabus 20.1

Question 14 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

440 Polymerisation — 9701/23/F/M/24/Q15(c) Syllabus 20.1

Question 15 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

441 Polymerisation — 9701/23/F/M/24/Q16(c) Syllabus 20.1

Question 16 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

442 Polymerisation — 9701/23/F/M/24/Q17(c) Syllabus 20.1

Question 17 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

443 Polymerisation — 9701/23/F/M/24/Q18(c) Syllabus 20.1

Question 18 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

444 Polymerisation — 9701/23/F/M/24/Q19(c) Syllabus 20.1

Question 19 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

445 Polymerisation — 9701/23/F/M/24/Q20(c) Syllabus 20.1

Question 20 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

Question Pack

TOPIC 21 — TOPIC 21: ORGANIC SYNTHESIS

446 Organic Synthesis — 9701/22/M/J/25/Q1(d) Syllabus 21.1

Question 1 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

447 Organic Synthesis — 9701/22/M/J/25/Q2(d) Syllabus 21.1

Question 2 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

448 Organic Synthesis — 9701/22/M/J/25/Q3(d) Syllabus 21.1

Question 3 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

449 Organic Synthesis — 9701/22/M/J/25/Q4(d) Syllabus 21.1

Question 4 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

450 Organic Synthesis — 9701/22/M/J/25/Q5(d) Syllabus 21.1

Question 5 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

451 Organic Synthesis — 9701/22/M/J/25/Q6(d) Syllabus 21.1

Question 6 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

452 Organic Synthesis — 9701/22/M/J/25/Q7(d) Syllabus 21.1

Question 7 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

453 Organic Synthesis — 9701/22/M/J/25/Q8(d) Syllabus 21.1

Question 8 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

454 Organic Synthesis — 9701/22/M/J/25/Q9(d) Syllabus 21.1

Question 9 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

455 Organic Synthesis — 9701/22/M/J/25/Q10(d) Syllabus 21.1

Question 10 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

Question Pack

456 Organic Synthesis — 9701/22/M/J/25/Q11(d) Syllabus 21.1

Question 11 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

457 Organic Synthesis — 9701/22/M/J/25/Q12(d) Syllabus 21.1

Question 12 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

458 Organic Synthesis — 9701/22/M/J/25/Q13(d) Syllabus 21.1

Question 13 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

459 Organic Synthesis — 9701/22/M/J/25/Q14(d) Syllabus 21.1

Question 14 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

460 Organic Synthesis — 9701/22/M/J/25/Q15(d) Syllabus 21.1

Question 15 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

461 Organic Synthesis — 9701/22/M/J/25/Q16(d) Syllabus 21.1

Question 16 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

462 Organic Synthesis — 9701/22/M/J/25/Q17(d) Syllabus 21.1

Question 17 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

463 Organic Synthesis — 9701/22/M/J/25/Q18(d) Syllabus 21.1

Question 18 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

464 Organic Synthesis — 9701/22/M/J/25/Q19(d) Syllabus 21.1

Question 19 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

465 Organic Synthesis — 9701/22/M/J/25/Q20(d) Syllabus 21.1

Question 20 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

Question Pack

TOPIC 22 — TOPIC 22: ANALYTICAL TECHNIQUES

466 Analytical Techniques — 9701/21/O/N/25/Q1(e) Syllabus 22.1

Question 1 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹.

[1]

467 Analytical Techniques — 9701/21/O/N/25/Q2(e) Syllabus 22.1

Question 2 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum.

[1]

468 Analytical Techniques — 9701/21/O/N/25/Q3(e) Syllabus 22.1

Question 3 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹.

[1]

469 Analytical Techniques — 9701/21/O/N/25/Q4(e) Syllabus 22.1

Question 4 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum.

[1]

470 Analytical Techniques — 9701/21/O/N/25/Q5(e) Syllabus 22.1

Question 5 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹.

[1]

471 Analytical Techniques — 9701/21/O/N/25/Q6(e) Syllabus 22.1

Question 6 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum.

[1]

472 Analytical Techniques — 9701/21/O/N/25/Q7(e) Syllabus 22.1

Question 7 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹.

[1]

473 Analytical Techniques — 9701/21/O/N/25/Q8(e) Syllabus 22.1

Question 8 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum.

[1]

474 Analytical Techniques — 9701/21/O/N/25/Q9(e) Syllabus 22.1

Question 9 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹.

[1]

475 Analytical Techniques — 9701/21/O/N/25/Q10(e) Syllabus 22.1

Question 10 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum.

[1]

Question Pack

476 Analytical Techniques — 9701/21/O/N/25/Q11(e) Syllabus 22.1

Question 11 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]

477 Analytical Techniques — 9701/21/O/N/25/Q12(e) Syllabus 22.1

Question 12 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum. [1]

478 Analytical Techniques — 9701/21/O/N/25/Q13(e) Syllabus 22.2

Question 13 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]

479 Analytical Techniques — 9701/21/O/N/25/Q14(e) Syllabus 22.2

Question 14 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum. [1]

480 Analytical Techniques — 9701/21/O/N/25/Q15(e) Syllabus 22.2

Question 15 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]

481 Analytical Techniques — 9701/21/O/N/25/Q16(e) Syllabus 22.2

Question 16 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum. [1]

482 Analytical Techniques — 9701/21/O/N/25/Q17(e) Syllabus 22.2

Question 17 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]

483 Analytical Techniques — 9701/21/O/N/25/Q18(e) Syllabus 22.2

Question 18 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum. [1]

484 Analytical Techniques — 9701/21/O/N/25/Q19(e) Syllabus 22.2

Question 19 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]

485 Analytical Techniques — 9701/21/O/N/25/Q20(e) Syllabus 22.2

Question 20 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum. [1]

Question Pack

486 Analytical Techniques — 9701/21/O/N/25/Q21(e) Syllabus 22.2

Question 21 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]

487 Analytical Techniques — 9701/21/O/N/25/Q22(e) Syllabus 22.2

Question 22 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum. [1]

488 Analytical Techniques — 9701/21/O/N/25/Q23(e) Syllabus 22.2

Question 23 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]

489 Analytical Techniques — 9701/21/O/N/25/Q24(e) Syllabus 22.2

Question 24 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum. [1]

490 Analytical Techniques — 9701/21/O/N/25/Q25(e) Syllabus 22.2

Question 25 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]

Mark Scheme

MARK SCHEME – WEEK 1

Q	MARKING POINTS	MARKS
1(a)	B (1)	1
2(a)	B (1)	1
3(a)	B (1)	1
4(a)	B (1)	1
5(a)	B (1)	1
6(a)	Atoms of the same element/same number of protons (1); with different numbers of neutrons/different mass number (1).	2
7(a)	Atoms of the same element/same number of protons (1); with different numbers of neutrons/different mass number (1).	2
8(a)	Atoms of the same element/same number of protons (1); with different numbers of neutrons/different mass number (1).	2
9(a)	Atoms of the same element/same number of protons (1); with different numbers of neutrons/different mass number (1).	2
10(a)	Atoms of the same element/same number of protons (1); with different numbers of neutrons/different mass number (1).	2
11(a)	Nuclear charge remains the same but electron being removed is from a more positive ion (1); so electrostatic attraction between nucleus and outer electron increases (1).	2
11(b)	Group 14 (or 4) due to large jump after 4th IE (1).	1
12(a)	Nuclear charge remains the same but electron being removed is from a more positive ion (1); so electrostatic attraction between nucleus and outer electron increases (1).	2
12(b)	Group 14 (or 4) due to large jump after 4th IE (1).	1
13(a)	Nuclear charge remains the same but electron being removed is from a more positive ion (1); so electrostatic attraction between nucleus and outer electron increases (1).	2
13(b)	Group 14 (or 4) due to large jump after 4th IE (1).	1
14(a)	Nuclear charge remains the same but electron being removed is from a more positive ion (1); so electrostatic attraction between nucleus and outer electron increases (1).	2
14(b)	Group 14 (or 4) due to large jump after 4th IE (1).	1
15(a)	Nuclear charge remains the same but electron being removed is from a more positive ion (1); so electrostatic attraction between nucleus and outer electron increases (1).	2
15(b)	Group 14 (or 4) due to large jump after 4th IE (1).	1
16(a)	A (1)	1
17(a)	A (1)	1
18(a)	A (1)	1

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
19(a)	A (1)	1
20(a)	A (1)	1
21(a)	Nuclear charge increases (1); shielding remains similar / electrons added to same shell (1); nuclear attraction for outer electrons increases (1).	3
22(a)	Nuclear charge increases (1); shielding remains similar / electrons added to same shell (1); nuclear attraction for outer electrons increases (1).	3
23(a)	Nuclear charge increases (1); shielding remains similar / electrons added to same shell (1); nuclear attraction for outer electrons increases (1).	3
24(a)	Nuclear charge increases (1); shielding remains similar / electrons added to same shell (1); nuclear attraction for outer electrons increases (1).	3
25(a)	Nuclear charge increases (1); shielding remains similar / electrons added to same shell (1); nuclear attraction for outer electrons increases (1).	3
26(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
27(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
28(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
29(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
30(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
31(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
32(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
33(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
34(a)	Moles of NaOH = $25.0 \times 0.100 / 1000 = 0.0025$ mol (1); Moles of H ₂ SO ₄ = $0.0025 / 2 = 0.00125$ mol (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0) = 0.0625$ mol dm ⁻³ (1).	3
35(a)	Moles of NaOH = $25.0 \times 0.100 / 1000 = 0.0025$ mol (1); Moles of H ₂ SO ₄ = $0.0025 / 2 = 0.00125$ mol (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0) = 0.0625$ mol dm ⁻³ (1).	3
36(a)	Moles of NaOH = $25.0 \times 0.100 / 1000 = 0.0025$ mol (1); Moles of H ₂ SO ₄ = $0.0025 / 2 = 0.00125$ mol (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0) = 0.0625$ mol dm ⁻³ (1).	3
37(a)	Moles of NaOH = $25.0 \times 0.100 / 1000 = 0.0025$ mol (1); Moles of H ₂ SO ₄ = $0.0025 / 2 = 0.00125$ mol (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0) = 0.0625$ mol dm ⁻³ (1).	3
38(a)	Moles of NaOH = $25.0 \times 0.100 / 1000 = 0.0025$ mol (1); Moles of H ₂ SO ₄ = $0.0025 / 2 = 0.00125$ mol (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0) = 0.0625$ mol dm ⁻³ (1).	3

Mark Scheme

MARK SCHEME – WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
39(a)	Moles of NaOH = $25.0 \times 0.100 / 1000 = 0.0025 \text{ mol}$ (1); Moles of H ₂ SO ₄ = $0.0025 / 2 = 0.00125 \text{ mol}$ (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0) = 0.0625 \text{ mol dm}^{-3}$ (1).	3
40(a)	Moles of NaOH = $25.0 \times 0.100 / 1000 = 0.0025 \text{ mol}$ (1); Moles of H ₂ SO ₄ = $0.0025 / 2 = 0.00125 \text{ mol}$ (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0) = 0.0625 \text{ mol dm}^{-3}$ (1).	3
41(a)	Moles of NaOH = $25.0 \times 0.100 / 1000 = 0.0025 \text{ mol}$ (1); Moles of H ₂ SO ₄ = $0.0025 / 2 = 0.00125 \text{ mol}$ (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0) = 0.0625 \text{ mol dm}^{-3}$ (1).	3
42(a)	Moles of NaOH = $25.0 \times 0.100 / 1000 = 0.0025 \text{ mol}$ (1); Moles of H ₂ SO ₄ = $0.0025 / 2 = 0.00125 \text{ mol}$ (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0) = 0.0625 \text{ mol dm}^{-3}$ (1).	3
43(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
44(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
45(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
46(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
47(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
48(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
49(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
50(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
51(a)	Trigonal pyramidal (1); 107 degrees (1).	2
52(a)	Trigonal pyramidal (1); 107 degrees (1).	2
53(a)	Trigonal pyramidal (1); 107 degrees (1).	2
54(a)	Trigonal pyramidal (1); 107 degrees (1).	2
55(a)	Trigonal pyramidal (1); 107 degrees (1).	2
56(a)	Trigonal pyramidal (1); 107 degrees (1).	2
57(a)	Trigonal pyramidal (1); 107 degrees (1).	2
58(a)	Trigonal pyramidal (1); 107 degrees (1).	2
59(a)	Two double covalent bonds (1); between C and O atoms (1).	2

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
60(a)	Two double covalent bonds (1); between C and O atoms (1).	2
61(a)	Two double covalent bonds (1); between C and O atoms (1).	2
62(a)	Two double covalent bonds (1); between C and O atoms (1).	2
63(a)	Two double covalent bonds (1); between C and O atoms (1).	2
64(a)	Two double covalent bonds (1); between C and O atoms (1).	2
65(a)	Two double covalent bonds (1); between C and O atoms (1).	2
66(a)	Two double covalent bonds (1); between C and O atoms (1).	2
67(a)	Two double covalent bonds (1); between C and O atoms (1).	2
68(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
69(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
70(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
71(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
72(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
73(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
74(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
75(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
76(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
77(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
78(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
79(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
80(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
81(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
82(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
83(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
84(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
85(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
86(a)	High temperature (1); Low pressure (1).	2
87(a)	High temperature (1); Low pressure (1).	2
88(a)	High temperature (1); Low pressure (1).	2
89(a)	High temperature (1); Low pressure (1).	2
90(a)	High temperature (1); Low pressure (1).	2
91(a)	High temperature (1); Low pressure (1).	2
92(a)	High temperature (1); Low pressure (1).	2
93(a)	High temperature (1); Low pressure (1).	2
94(a)	High temperature (1); Low pressure (1).	2
95(a)	High temperature (1); Low pressure (1).	2
96(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
97(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
98(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
99(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
100(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
101(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
102(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
103(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
104(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
105(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
106(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
107(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
108(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
109(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
110(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
111(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
112(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
113(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
114(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
115(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
116(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
117(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
118(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
119(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
120(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
121(a)	C (1)	1
122(a)	C (1)	1
123(a)	C (1)	1
124(a)	C (1)	1
125(a)	C (1)	1
126(a)	C (1)	1
127(a)	C (1)	1
128(a)	C (1)	1
129(a)	$Q = It = 18 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
130(a)	$Q = It = 20 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
131(a)	$Q = It = 22 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
132(a)	$Q = It = 24 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
133(a)	$Q = It = 26 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
134(a)	$Q = It = 28 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
135(a)	$Q = It = 30 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
136(a)	$Q = It = 32 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
137(a)	$Q = It = 34 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
138(a)	$Q = It = 36 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
139(a)	$Q = It = 38 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
140(a)	$Q = It = 40 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
141(a)	B (1)	1
142(a)	B (1)	1
143(a)	B (1)	1

Mark Scheme
MARK SCHEME – WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
144(a)	B (1)	1
145(a)	B (1)	1
146(a)	B (1)	1
147(a)	B (1)	1
148(a)	B (1)	1
149(a)	B (1)	1
150(a)	B (1)	1
151(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
151(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
152(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
152(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
153(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
153(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
154(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
154(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
155(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
155(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
156(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
156(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
157(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
157(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
158(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
158(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
159(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
159(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
160(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
160(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
161(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
161(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
162(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
162(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
163(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
163(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
164(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
164(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
165(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
165(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
166(a)	B (1)	1
167(a)	B (1)	1
168(a)	B (1)	1
169(a)	B (1)	1
170(a)	B (1)	1
171(a)	B (1)	1
172(a)	B (1)	1
173(a)	B (1)	1
174(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
175(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
176(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
177(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
178(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
179(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
180(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
181(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
182(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
183(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
184(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
185(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
186(a)	C (1)	1
187(a)	C (1)	1
188(a)	C (1)	1
189(a)	C (1)	1
190(a)	C (1)	1
191(a)	C (1)	1
192(a)	C (1)	1
193(a)	C (1)	1

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
194(a)	C (1)	1
195(a)	C (1)	1
196(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
196(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
197(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
197(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
198(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
198(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
199(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
199(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
200(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
200(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
201(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
201(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
202(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
202(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
203(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
203(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
204(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
204(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
205(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
205(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
206(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
206(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
207(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
207(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
208(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
208(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
209(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
209(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
210(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
210(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
211(a)	Pale green / apple green (1).	1
212(a)	Stability increases down the group (1); cation radius increases so less polarising power (1); less distortion of carbonate ion (1).	3
213(a)	$2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$ (1 for species, 1 for balancing).	2
214(a)	Barium is more reactive than magnesium (1). Barium reacts vigorously with cold water whereas Mg reacts very slowly (1).	2
215(a)	Solubility increases down the group (1).	1
216(a)	Pale green / apple green (1).	1
217(a)	Stability increases down the group (1); cation radius increases so less polarising power (1); less distortion of carbonate ion (1).	3
218(a)	$2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$ (1 for species, 1 for balancing).	2
219(a)	Barium is more reactive than magnesium (1). Barium reacts vigorously with cold water whereas Mg reacts very slowly (1).	2
220(a)	Solubility increases down the group (1).	1
221(a)	Pale green / apple green (1).	1
222(a)	Stability increases down the group (1); cation radius increases so less polarising power (1); less distortion of carbonate ion (1).	3

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
223(a)	$2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$ (1 for species, 1 for balancing).	2
224(a)	Barium is more reactive than magnesium (1). Barium reacts vigorously with cold water whereas Mg reacts very slowly (1).	2
225(a)	Solubility increases down the group (1).	1
226(a)	Pale green / apple green (1).	1
227(a)	Stability increases down the group (1); cation radius increases so less polarising power (1); less distortion of carbonate ion (1).	3
228(a)	$2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$ (1 for species, 1 for balancing).	2
229(a)	Barium is more reactive than magnesium (1). Barium reacts vigorously with cold water whereas Mg reacts very slowly (1).	2
230(a)	Solubility increases down the group (1).	1
231(a)	AgCl forms a soluble complex ion $[\text{Ag}(\text{NH}_3)_2]^+$ (1); AgI is less soluble so does not dissolve in dilute NH_3 (1).	2
232(a)	$\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$ (1 for species, 1 for balancing).	2
233(a)	AgNO ₃ gives coloured precipitates (1); ammonia tests solubility of precipitates to differentiate between Cl ⁻ , Br ⁻ , I ⁻ (1).	2
234(a)	Chlorine kills bacteria / microorganisms (1); making water safe to drink (1).	2
235(a)	Boiling points increase down the group (1); due to more electrons so stronger van der Waals' forces (1).	2
236(a)	AgCl forms a soluble complex ion $[\text{Ag}(\text{NH}_3)_2]^+$ (1); AgI is less soluble so does not dissolve in dilute NH_3 (1).	2
237(a)	$\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$ (1 for species, 1 for balancing).	2
238(a)	AgNO ₃ gives coloured precipitates (1); ammonia tests solubility of precipitates to differentiate between Cl ⁻ , Br ⁻ , I ⁻ (1).	2
239(a)	Chlorine kills bacteria / microorganisms (1); making water safe to drink (1).	2
240(a)	Boiling points increase down the group (1); due to more electrons so stronger van der Waals' forces (1).	2
241(a)	AgCl forms a soluble complex ion $[\text{Ag}(\text{NH}_3)_2]^+$ (1); AgI is less soluble so does not dissolve in dilute NH_3 (1).	2
242(a)	$\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$ (1 for species, 1 for balancing).	2
243(a)	AgNO ₃ gives coloured precipitates (1); ammonia tests solubility of precipitates to differentiate between Cl ⁻ , Br ⁻ , I ⁻ (1).	2
244(a)	Chlorine kills bacteria / microorganisms (1); making water safe to drink (1).	2

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
245(a)	Boiling points increase down the group (1); due to more electrons so stronger van der Waals' forces (1).	2
246(a)	AgCl forms a soluble complex ion $[\text{Ag}(\text{NH}_3)_2]^+$ (1); AgI is less soluble so does not dissolve in dilute NH_3 (1).	2
247(a)	$\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$ (1 for species, 1 for balancing).	2
248(a)	AgNO_3 gives coloured precipitates (1); ammonia tests solubility of precipitates to differentiate between Cl^- , Br^- , I^- (1).	2
249(a)	Chlorine kills bacteria / microorganisms (1); making water safe to drink (1).	2
250(a)	Boiling points increase down the group (1); due to more electrons so stronger van der Waals' forces (1).	2
251(a)	AgCl forms a soluble complex ion $[\text{Ag}(\text{NH}_3)_2]^+$ (1); AgI is less soluble so does not dissolve in dilute NH_3 (1).	2
252(a)	$\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$ (1 for species, 1 for balancing).	2
253(a)	AgNO_3 gives coloured precipitates (1); ammonia tests solubility of precipitates to differentiate between Cl^- , Br^- , I^- (1).	2
254(a)	Chlorine kills bacteria / microorganisms (1); making water safe to drink (1).	2
255(a)	Boiling points increase down the group (1); due to more electrons so stronger van der Waals' forces (1).	2
256(a)	Forms acid rain (1); which lowers pH of lakes / damages buildings (1).	2
257(a)	$2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_2 + \text{HNO}_3$ (1).	1
258(a)	$2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$ (1); Pt/Pd/Rh catalyst lowers activation energy (1).	2
259(a)	Volcanoes (1).	1
260(a)	High temperature in engine (1); causes nitrogen and oxygen from the air to react (1).	2
261(a)	Forms acid rain (1); which lowers pH of lakes / damages buildings (1).	2
262(a)	$2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_2 + \text{HNO}_3$ (1).	1
263(a)	$2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$ (1); Pt/Pd/Rh catalyst lowers activation energy (1).	2
264(a)	Volcanoes (1).	1
265(a)	High temperature in engine (1); causes nitrogen and oxygen from the air to react (1).	2
266(a)	Forms acid rain (1); which lowers pH of lakes / damages buildings (1).	2
267(a)	$2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_2 + \text{HNO}_3$ (1).	1

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
268(a)	$2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$ (1); Pt/Pd/Rh catalyst lowers activation energy (1).	2
269(a)	Volcanoes (1).	1
270(a)	High temperature in engine (1); causes nitrogen and oxygen from the air to react (1).	2
271(a)	Forms acid rain (1); which lowers pH of lakes / damages buildings (1).	2
272(a)	$2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_2 + \text{HNO}_3$ (1).	1
273(a)	$2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$ (1); Pt/Pd/Rh catalyst lowers activation energy (1).	2
274(a)	Volcanoes (1).	1
275(a)	High temperature in engine (1); causes nitrogen and oxygen from the air to react (1).	2
276(a)	Structural isomers have different connectivity of atoms (1); stereoisomers have same connectivity but different spatial arrangement (1).	2
277(a)	Compare atomic number of atoms attached to double bond carbons (1); higher atomic number has higher priority (1).	2
278(a)	Amine / NH_2 (1); Carboxylic acid / COOH (1).	2
279(a)	Oxidised to aldehyde (1); then to carboxylic acid (1); using acidified potassium dichromate(VI) (1).	3
280(a)	butan-2-ol (1).	1
281(a)	Structural isomers have different connectivity of atoms (1); stereoisomers have same connectivity but different spatial arrangement (1).	2
282(a)	Compare atomic number of atoms attached to double bond carbons (1); higher atomic number has higher priority (1).	2
283(a)	Amine / NH_2 (1); Carboxylic acid / COOH (1).	2
284(a)	Oxidised to aldehyde (1); then to carboxylic acid (1); using acidified potassium dichromate(VI) (1).	3
285(a)	butan-2-ol (1).	1
286(a)	Structural isomers have different connectivity of atoms (1); stereoisomers have same connectivity but different spatial arrangement (1).	2
287(a)	Compare atomic number of atoms attached to double bond carbons (1); higher atomic number has higher priority (1).	2
288(a)	Amine / NH_2 (1); Carboxylic acid / COOH (1).	2
289(a)	Oxidised to aldehyde (1); then to carboxylic acid (1); using acidified potassium dichromate(VI) (1).	3

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
290(a)	butan-2-ol (1).	1
291(a)	Structural isomers have different connectivity of atoms (1); stereoisomers have same connectivity but different spatial arrangement (1).	2
292(a)	Compare atomic number of atoms attached to double bond carbons (1); higher atomic number has higher priority (1).	2
293(a)	Amine / NH ₂ (1); Carboxylic acid / COOH (1).	2
294(a)	Oxidised to aldehyde (1); then to carboxylic acid (1); using acidified potassium dichromate(VI) (1).	3
295(a)	butan-2-ol (1).	1
296(a)	Structural isomers have different connectivity of atoms (1); stereoisomers have same connectivity but different spatial arrangement (1).	2
297(a)	Compare atomic number of atoms attached to double bond carbons (1); higher atomic number has higher priority (1).	2
298(a)	Amine / NH ₂ (1); Carboxylic acid / COOH (1).	2
299(a)	Oxidised to aldehyde (1); then to carboxylic acid (1); using acidified potassium dichromate(VI) (1).	3
300(a)	butan-2-ol (1).	1
301(a)	Free radical substitution (1); UV light / sunlight / high temperature (1).	2
301(b)	Initiation: Cl ₂ → 2Cl. (1); Propagation: Cl. + ethane → alkyl. + HCl (1); alkyl. + Cl ₂ → chloroalkane + Cl. (1); Termination: Cl. + Cl. → Cl ₂ or two radicals combining (1).	4
302(a)	Free radical substitution (1); UV light / sunlight / high temperature (1).	2
302(b)	Initiation: Cl ₂ → 2Cl. (1); Propagation: Cl. + propane → alkyl. + HCl (1); alkyl. + Cl ₂ → chloroalkane + Cl. (1); Termination: Cl. + Cl. → Cl ₂ or two radicals combining (1).	4
303(a)	Free radical substitution (1); UV light / sunlight / high temperature (1).	2
303(b)	Initiation: Cl ₂ → 2Cl. (1); Propagation: Cl. + butane → alkyl. + HCl (1); alkyl. + Cl ₂ → chloroalkane + Cl. (1); Termination: Cl. + Cl. → Cl ₂ or two radicals combining (1).	4
304(a)	Free radical substitution (1); UV light / sunlight / high temperature (1).	2
304(b)	Initiation: Cl ₂ → 2Cl. (1); Propagation: Cl. + pentane → alkyl. + HCl (1); alkyl. + Cl ₂ → chloroalkane + Cl. (1); Termination: Cl. + Cl. → Cl ₂ or two radicals combining (1).	4
305(a)	Free radical substitution (1); UV light / sunlight / high temperature (1).	2
305(b)	Initiation: Cl ₂ → 2Cl. (1); Propagation: Cl. + hexane → alkyl. + HCl (1); alkyl. + Cl ₂ → chloroalkane + Cl. (1); Termination: Cl. + Cl. → Cl ₂ or two radicals combining (1).	4
306(a)	Free radical substitution (1); UV light / sunlight / high temperature (1).	2

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
306(b)	Initiation: $\text{Cl}_2 \rightarrow 2\text{Cl} \cdot$ (1); Propagation: $\text{Cl} \cdot + \text{ethane} \rightarrow \text{alkyl} \cdot + \text{HCl}$ (1); $\text{alkyl} \cdot + \text{Cl}_2 \rightarrow \text{chloroalkane} + \text{Cl} \cdot$ (1); Termination: $\text{Cl} \cdot + \text{Cl} \cdot \rightarrow \text{Cl}_2$ or two radicals combining (1).	4
307(a)	Free radical substitution (1); UV light / sunlight / high temperature (1).	2
307(b)	Initiation: $\text{Cl}_2 \rightarrow 2\text{Cl} \cdot$ (1); Propagation: $\text{Cl} \cdot + \text{propane} \rightarrow \text{alkyl} \cdot + \text{HCl}$ (1); $\text{alkyl} \cdot + \text{Cl}_2 \rightarrow \text{chloroalkane} + \text{Cl} \cdot$ (1); Termination: $\text{Cl} \cdot + \text{Cl} \cdot \rightarrow \text{Cl}_2$ or two radicals combining (1).	4
308(a)	Free radical substitution (1); UV light / sunlight / high temperature (1).	2
308(b)	Initiation: $\text{Cl}_2 \rightarrow 2\text{Cl} \cdot$ (1); Propagation: $\text{Cl} \cdot + \text{butane} \rightarrow \text{alkyl} \cdot + \text{HCl}$ (1); $\text{alkyl} \cdot + \text{Cl}_2 \rightarrow \text{chloroalkane} + \text{Cl} \cdot$ (1); Termination: $\text{Cl} \cdot + \text{Cl} \cdot \rightarrow \text{Cl}_2$ or two radicals combining (1).	4
309(a)	Free radical substitution (1); UV light / sunlight / high temperature (1).	2
309(b)	Initiation: $\text{Cl}_2 \rightarrow 2\text{Cl} \cdot$ (1); Propagation: $\text{Cl} \cdot + \text{pentane} \rightarrow \text{alkyl} \cdot + \text{HCl}$ (1); $\text{alkyl} \cdot + \text{Cl}_2 \rightarrow \text{chloroalkane} + \text{Cl} \cdot$ (1); Termination: $\text{Cl} \cdot + \text{Cl} \cdot \rightarrow \text{Cl}_2$ or two radicals combining (1).	4
310(a)	Free radical substitution (1); UV light / sunlight / high temperature (1).	2
310(b)	Initiation: $\text{Cl}_2 \rightarrow 2\text{Cl} \cdot$ (1); Propagation: $\text{Cl} \cdot + \text{hexane} \rightarrow \text{alkyl} \cdot + \text{HCl}$ (1); $\text{alkyl} \cdot + \text{Cl}_2 \rightarrow \text{chloroalkane} + \text{Cl} \cdot$ (1); Termination: $\text{Cl} \cdot + \text{Cl} \cdot \rightarrow \text{Cl}_2$ or two radicals combining (1).	4
311(a)	Free radical substitution (1); UV light / sunlight / high temperature (1).	2
311(b)	Initiation: $\text{Cl}_2 \rightarrow 2\text{Cl} \cdot$ (1); Propagation: $\text{Cl} \cdot + \text{ethane} \rightarrow \text{alkyl} \cdot + \text{HCl}$ (1); $\text{alkyl} \cdot + \text{Cl}_2 \rightarrow \text{chloroalkane} + \text{Cl} \cdot$ (1); Termination: $\text{Cl} \cdot + \text{Cl} \cdot \rightarrow \text{Cl}_2$ or two radicals combining (1).	4
312(a)	Free radical substitution (1); UV light / sunlight / high temperature (1).	2
312(b)	Initiation: $\text{Cl}_2 \rightarrow 2\text{Cl} \cdot$ (1); Propagation: $\text{Cl} \cdot + \text{propane} \rightarrow \text{alkyl} \cdot + \text{HCl}$ (1); $\text{alkyl} \cdot + \text{Cl}_2 \rightarrow \text{chloroalkane} + \text{Cl} \cdot$ (1); Termination: $\text{Cl} \cdot + \text{Cl} \cdot \rightarrow \text{Cl}_2$ or two radicals combining (1).	4
313(a)	Free radical substitution (1); UV light / sunlight / high temperature (1).	2
313(b)	Initiation: $\text{Cl}_2 \rightarrow 2\text{Cl} \cdot$ (1); Propagation: $\text{Cl} \cdot + \text{butane} \rightarrow \text{alkyl} \cdot + \text{HCl}$ (1); $\text{alkyl} \cdot + \text{Cl}_2 \rightarrow \text{chloroalkane} + \text{Cl} \cdot$ (1); Termination: $\text{Cl} \cdot + \text{Cl} \cdot \rightarrow \text{Cl}_2$ or two radicals combining (1).	4
314(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
314(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
315(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
315(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
316(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
316(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
317(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
317(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
318(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
318(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
319(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
319(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
320(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
320(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
321(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
321(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
322(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
322(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
323(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
323(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
324(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
324(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
325(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
325(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
326(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
326(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
326(c)	Primary undergo SN ₂ , tertiary undergo SN ₁ (1); SN ₁ involves carbocation intermediate (1); SN ₂ involves a transition state / one step mechanism (1).	3
327(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
327(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
327(c)	Primary undergo SN ₂ , tertiary undergo SN ₁ (1); SN ₁ involves carbocation intermediate (1); SN ₂ involves a transition state / one step mechanism (1).	3
328(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
328(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
328(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
329(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
329(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
329(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
330(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
330(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
330(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
331(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
331(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
331(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
332(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
332(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
332(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
333(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
333(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
333(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
334(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
334(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
334(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
335(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
335(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
335(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
336(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
336(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
336(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
337(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
337(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
337(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
338(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
338(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
338(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
339(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
339(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
339(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
340(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
340(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
340(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
341(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
341(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
341(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
342(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
342(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
342(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
343(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
343(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
343(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
344(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
344(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
344(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
345(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
345(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
345(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
346(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C-OH carbon (1).	2
346(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
346(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
347(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C-OH carbon (1).	2
347(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
347(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
348(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C-OH carbon (1).	2
348(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
348(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
349(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C-OH carbon (1).	2
349(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
349(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
350(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C-OH carbon (1).	2
350(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
350(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
351(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C-OH carbon (1).	2
351(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
351(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
352(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
352(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
352(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
353(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
353(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
353(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
354(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
354(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
354(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
355(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
355(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
355(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
356(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
356(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
356(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
357(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
357(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
357(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
358(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
358(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
358(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
359(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
359(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
359(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
360(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
360(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3

Mark Scheme

MARK SCHEME – WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
360(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
361(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
361(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
361(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
362(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
362(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
362(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
363(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
363(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
363(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
364(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
364(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
364(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
365(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
365(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
365(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
366(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
366(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
366(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
367(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
367(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
367(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
368(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
368(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
368(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
369(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
369(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
369(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
370(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
370(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
370(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
371(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
371(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
371(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
372(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
372(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
372(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
373(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
373(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
373(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
374(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
374(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
374(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
375(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
375(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
375(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
376(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
376(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
376(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
377(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
377(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
377(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
378(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
378(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
378(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
379(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
379(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
379(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
380(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
380(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
380(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
381(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
381(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
381(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
382(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
382(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
382(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
383(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
383(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
383(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
384(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
384(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
384(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
385(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
385(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
385(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
386(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
387(a)	B (1)	1
388(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
389(a)	B (1)	1
390(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
391(a)	B (1)	1
392(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
393(a)	B (1)	1
394(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
395(a)	B (1)	1
396(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
397(a)	B (1)	1
398(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
399(a)	B (1)	1
400(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
401(a)	B (1)	1
402(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
403(a)	B (1)	1

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
404(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
405(a)	B (1)	1
406(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
407(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
408(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
409(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
410(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
411(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
412(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
413(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
414(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
415(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
416(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
417(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
418(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
419(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
420(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
421(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
422(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
423(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
424(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
425(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
426(a)	Correct repeat unit with continuation bonds (1)	1

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
427(a)	Correct repeat unit with continuation bonds (1)	1
428(a)	Correct repeat unit with continuation bonds (1)	1
429(a)	Correct repeat unit with continuation bonds (1)	1
430(a)	Correct repeat unit with continuation bonds (1)	1
431(a)	Correct repeat unit with continuation bonds (1)	1
432(a)	Correct repeat unit with continuation bonds (1)	1
433(a)	Correct repeat unit with continuation bonds (1)	1
434(a)	Correct repeat unit with continuation bonds (1)	1
435(a)	Correct repeat unit with continuation bonds (1)	1
436(a)	Correct repeat unit with continuation bonds (1)	1
437(a)	Correct repeat unit with continuation bonds (1)	1
438(a)	Correct repeat unit with continuation bonds (1)	1
439(a)	Correct repeat unit with continuation bonds (1)	1
440(a)	Correct repeat unit with continuation bonds (1)	1
441(a)	Correct repeat unit with continuation bonds (1)	1
442(a)	Correct repeat unit with continuation bonds (1)	1
443(a)	Correct repeat unit with continuation bonds (1)	1
444(a)	Correct repeat unit with continuation bonds (1)	1
445(a)	Correct repeat unit with continuation bonds (1)	1
446(a)	Correct structure (1); correct reagents (1)	2
447(a)	Correct structure (1); correct reagents (1)	2
448(a)	Correct structure (1); correct reagents (1)	2
449(a)	Correct structure (1); correct reagents (1)	2

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
450(a)	Correct structure (1); correct reagents (1)	2
451(a)	Correct structure (1); correct reagents (1)	2
452(a)	Correct structure (1); correct reagents (1)	2
453(a)	Correct structure (1); correct reagents (1)	2
454(a)	Correct structure (1); correct reagents (1)	2
455(a)	Correct structure (1); correct reagents (1)	2
456(a)	Correct structure (1); correct reagents (1)	2
457(a)	Correct structure (1); correct reagents (1)	2
458(a)	Correct structure (1); correct reagents (1)	2
459(a)	Correct structure (1); correct reagents (1)	2
460(a)	Correct structure (1); correct reagents (1)	2
461(a)	Correct structure (1); correct reagents (1)	2
462(a)	Correct structure (1); correct reagents (1)	2
463(a)	Correct structure (1); correct reagents (1)	2
464(a)	Correct structure (1); correct reagents (1)	2
465(a)	Correct structure (1); correct reagents (1)	2
466(a)	O-H / alcohol (1)	1
467(a)	Gives the relative molecular mass of the compound (1)	1
468(a)	O-H / alcohol (1)	1
469(a)	Gives the relative molecular mass of the compound (1)	1
470(a)	O-H / alcohol (1)	1
471(a)	Gives the relative molecular mass of the compound (1)	1
472(a)	O-H / alcohol (1)	1

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
473(a)	Gives the relative molecular mass of the compound (1)	1
474(a)	O-H / alcohol (1)	1
475(a)	Gives the relative molecular mass of the compound (1)	1
476(a)	O-H / alcohol (1)	1
477(a)	Gives the relative molecular mass of the compound (1)	1
478(a)	O-H / alcohol (1)	1
479(a)	Gives the relative molecular mass of the compound (1)	1
480(a)	O-H / alcohol (1)	1
481(a)	Gives the relative molecular mass of the compound (1)	1
482(a)	O-H / alcohol (1)	1
483(a)	Gives the relative molecular mass of the compound (1)	1
484(a)	O-H / alcohol (1)	1
485(a)	Gives the relative molecular mass of the compound (1)	1
486(a)	O-H / alcohol (1)	1
487(a)	Gives the relative molecular mass of the compound (1)	1
488(a)	O-H / alcohol (1)	1
489(a)	Gives the relative molecular mass of the compound (1)	1
490(a)	O-H / alcohol (1)	1